Modeling of kilonova light curves from neutron star mergers and possible constraints on nuclear physics

by

Diego Fernando Hernández Porras

Submitted to the Department of Physics
in partial fulfillment of the requirements for the degree of

MASTER OF SCIENCE IN PHYSICS

at the

UNIVERSIDAD DE LOS ANDES

December 2026

© 2026 Diego Fernando Hernández Porras. All rights reserved.

The author hereby grants to Universidad de los Andes a nonexclusive, worldwide, irrevocable, royalty-free license to exercise any and all rights under copyright, including to reproduce, preserve, distribute and publicly display copies of the thesis, or release the thesis under an open-access license.

Authored by: Diego Fernando Hernández Porras
Department of Physics
December, 2026

Certified by: Neelima Govind Kelkar
Professor of Physics, Thesis Supervisor

Accepted by: None
Departament of Physics
none

Modeling of kilonova light curves from neutron star mergers and possible constraints on nuclear physics

by

Diego Fernando Hernández Porras

Submitted to the Department of Physics
on December, 2026 in partial fulfillment of the requirements for the degree of

MASTER OF SCIENCE IN PHYSICS

ABSTRACT

Neutron stars are compact objects whose hydrostatic equilibrium is determined by the equation of state (EOS) of the dense matter in their interiors, which makes them natural laboratories for nuclear physics at densities inaccessible to terrestrial experiments. When two neutron stars merge, part of their mass is ejected and, as demonstrated by the gravitational-wave event GW170817 and its associated kilonova, this ejected material is a site of rapid neutron capture (r-process) nucleosynthesis. The yield of heavy nuclei in the ejecta depends on the properties of neutron-rich nuclei far from stability, including nuclear masses, decay rates, neutron-capture rates, and fission properties, among others, many of which are unknown. Since nuclear physics plays a crucial role in the evolution of observables, understanding the impact of relying on theoretical nuclear physics results is essential to correctly interpret current and future multimessenger observations.

This work presents a detailed analysis of the impact of different theoretical α- and β^--decay rates on the kilonova evolution. The nucleosynthesis was calculated using the nuclear reaction network WinNet across an ensemble of initial thermodynamic conditions parametrized by the initial electron fraction ($Y_{e,0}$), the expansion timescale ($\tau_{\rm exp}$) and the initial entropy per baryon (s_0). The kilonova light curves were calculated using current semi-analytical models based on radiative-transfer simulations. The results show that the heating rate from radioactive decays deviates from the accepted power-law behavior depending on the β^--decay prescription consistently across the initial conditions studied. While β^- decay dominates until about 10 days post-merger, α decay takes over at later times. The dominance with about 2–3 orders of magnitude higher heating rates than others can be attributed mostly to nuclei with $A \geq 240$. The set of nuclei responsible for such behavior was identified and the description of their decay chains is presented.

The results on the impact of theoretical rates described above are then tested in a merger simulation where the resulting kilonova is now described as a superposition of different ejecta components[1]. The amount of ejected mass and its velocity are extracted from a general-relativistic simulation which was computed using the Einstein Toolkit framework.

Thesis supervisor: Neelima Govind Kelkar
Title: Professor of Physics

[1]This paragraph may change as results from GR simulations are being obtained

Acknowledgments

Este trabajo respresenta el final de una de las etapas mas bonitas de mi vida. Llevaré con mucho cariño el recuerdo de todos mis amigos a donde me lleve la motivación.

Contents

1 Introduction **6**
- 1.1 Kilonovae in a nutshell . . . 6
- 1.2 A brief history of the kilonova . . . 7
- 1.3 Modern models of kilonovae and observations . . . 8
- 1.4 Objectives and thesis structure . . . 10

2 From Neutron Stars to Kilonovae **12**
- 2.1 The Neutron Star . . . 12
- 2.2 Formation of binary neutron stars . . . 13
- 2.3 Gravitational-wave-driven inspiral and the maximum separation . . . 15
- 2.4 Mass ejection: tidal, shock-driven, and disk-wind ejecta . . . 16
  - 2.4.1 Dynamical ejecta . . . 17
  - 2.4.2 Post-merger ejecta . . . 17
- 2.5 Weak interactions and the electron fraction of the ejecta . . . 18
- 2.6 Electromagnetic counterparts . . . 19
  - 2.6.1 GRBs . . . 20
  - 2.6.2 Kilonovae . . . 20

3 r-Process Nucleosynthesis **24**
- 3.1 Abundances, reaction rates and the network equation . . . 24
  - 3.1.1 Abundances . . . 24
  - 3.1.2 Reaction rates . . . 25
  - 3.1.3 The network equation . . . 27
  - 3.1.4 Detailed balance . . . 28
  - 3.1.5 The network during the r-process . . . 29
- 3.2 Nuclear statistical equilibrium and freeze-out . . . 30
  - 3.2.1 The Saha equation . . . 30
  - 3.2.2 Composition in NSE: binding energy versus entropy . . . 31
  - 3.2.3 Quasi-equilibrium and freeze-out . . . 32
- 3.3 The r-process: neutron-to-seed ratio, reaction path and fission cycling . . . 33
  - 3.3.1 Neutron-to-seed ratio . . . 33
  - 3.3.2 The reaction path and waiting points . . . 34
  - 3.3.3 Freeze-out and decay to stability . . . 35
  - 3.3.4 Fission . . . 35
- 3.4 Radioactive energy release, decay chains and self-heating . . . 35

3.4.1 Nuclear energy generation and the entropy equation . . . 36
3.4.2 Energy partition and neutrino losses . . . 37
3.4.3 Self-heating . . . 38
3.4.4 Decay chains . . . 39

4 Kilonova Light Curves **41**
4.1 Li-Paczyński model . . . 41
4.2 Radiative transfer and the diffusion regime . . . 41
4.3 Opacities of r-process matter . . . 41
4.4 Light-curve modeling . . . 41

5 Methodology on the impact of theoretical decay rates across thermodynamics trajectories **42**
5.1 Binary neutron-star simulation . . . 42
5.2 From simulation to trajectories: ejecta extraction and expansion . . . 42
5.3 Initial termodinamic trajectories treatment . . . 42
5.4 Nuclear reaction network setup and theoretical reaction reates . . . 42
5.5 Modified Li-Paczyński model . . . 42

6 Results **43**
6.1 Nucleosynthesis and heating rates . . . 43
6.1.1 Relevant α and β . . . 43
6.2 Kilonova light curves . . . 43
6.3 Detectability with current and future telescopes . . . 43

7 Discussion **44**
7.1 Comparison with previous studies . . . 44
7.2 Limitations and model assumptions . . . 44

8 Conclusions and Outlook **45**

A Quadrupole and L_{GW} **46**

B Derivation of the Saha equation of nuclear statistical equilibrium **48**
B.1 Ideal non-degenerate gas of particles with internal structure . . . 48
B.2 The Saha equation for number densities . . . 49
B.3 From number densities to abundances . . . 50

References 51

Chapter 1

Introduction

Binary neutron star mergers (BNSMs) are among the most energetic events in the Universe, releasing tremendous amounts of energy through gravitational waves, neutrinos, and electromagnetic radiation. During and after the merger of two neutron stars, a fraction of their mass is ejected through tidal interactions, shocks, and disk winds. The ejected material is expected to consist of extremely neutron-rich matter, within which rapid neutron captures (r-process) lead to the formation of heavy nuclei far from stability. Consequently, these mergers are considered one of the primary sites for the synthesis of the heaviest elements in the Universe [1, 2]. As the freshly synthesized nuclei decay toward stability, their radioactive energy release drives the thermodynamic evolution of the ejecta and produces a thermal electromagnetic transient [3]. Such a transient is called a kilonova and will be the main topic of this thesis, since it encodes crucial information about the physics of the merger, leading to the abundance pattern of the elements formed in the r-process nucleosynthesis. Comparison of the observed light curves with models also allows us to constrain and understand the underlying nuclear physics responsible for powering the kilonovae.

1.1 Kilonovae in a nutshell

Kilonovae are closely related to the nuclear and atomic phenomena inside the ejecta. As heavy nuclei decay toward stability, part of the energy is released in the form of high energy photons according to the specific transitions inside the nuclei. In addition, decay products such as α and β particles, which also carry part of the decay energy, interact with surrounding charged particles primarily through Coulomb interaction, producing photons that carry away part of that energy. Effectively, the relevant particles will be photons.

At early stages the density is still high, so photons cannot easily escape and interact with several charged particles before reaching the surface. As photons interact with matter, their energy spectrum approaches that of a blackbody in a process called thermalization, and the same process applies to all the other decay products [4]. The emission at such stages is well described as a blackbody with a temperature evolution governed by the heating rate from radioactive decays. As the density decreases, bound states such as atoms form in a process called recombination. After this point the thermalization of the photons depends on the emission and absorption spectra of heavy elements. As a consequence, kilonova light curves

encapsulate information about the abundance of certain nuclei with specific emission and absorption lines. For example, the presence of lanthanides increases the opacity of the ejected material by an order of magnitude, so a kilonova with lanthanides will thermalize more efficiently and develop a long-lived thermal spectrum [5]. Since the abundances of lanthanides and of the rest of nuclei are determined by nuclear physics inputs, such as reaction rates, decay rates and masses, a detailed treatment of their uncertainties is needed for the correct interpretation of current and future observations. This is where this research work sits, as developed later in the text.

In the following sections, we will first give the reader a brief historical introduction to the kilonova. Then we will present modern kilonova models and their limitations. By the end of this chapter, we will state the objective of the research, explain the methodology, and describe the structure of this thesis.

1.2 A brief history of the kilonova

Until the last decade, the astrophysical origin of roughly half of the nuclei heavier than iron remained one of the major open problems of nuclear astrophysics. Big Bang nucleosynthesis accounts for the primordial abundances of the lightest nuclides (D, ^{3}He, ^{4}He and ^{7}Li), while hydrostatic and explosive burning in stars and supernovae produces the elements from carbon up to the iron peak and some heavier elements [6, 7]. The issue was that, beyond iron, the binding energy per nucleon decreases and Coulomb barriers grow, so heavy nuclei are built mainly by neutron captures rather than by charged particle fusion. Two main channels were known [8]. The first channel is the slow (s) process, in which neutron captures are slower than β-decays. The second is the rapid (r) process, in which captures are much faster than β-decays and drive matter far to the neutron-rich side of stability. The s-process is now well understood and operates mainly in asymptotic giant branch (AGB) stars and, for the lighter heavy elements, in massive stars [9]. Thanks to the study of s-process predictions and the later comparison with observations it was possible to identify the solar system r-abundance [10]. With the solar system r-abundances and the conditions required by the r-process understood it was possible to constrain the possible astrophysical sites. Those conditions are neutron densities $n_n \gtrsim 10^{20}\,\mathrm{cm}^{-3}$, or equivalently suitable combinations of electron fraction, entropy and expansion timescale in the ejecta [11]. However, the site where these conditions occur remained elusive.

For several decades the favored candidate was the core-collapse supernova, in particular the neutrino-driven wind from the newborn neutron star [12]. Modern simulations, however, find this wind neither neutron-rich nor entropic enough to produce the heaviest r-process nuclei [13], so ordinary supernovae contribute at most a weak r-process, producing elements up to about Sr–Ag. Observations point in the same direction. The large star-to-star scatter of europium in metal-poor stars [14], the strong r-process enrichment of the ultra-faint dwarf galaxy Reticulum II [15], and the low abundance of live ^{244}Pu in deep-sea sediments [16] all indicate that the heavy r-process elements are made in rare events with large yields, rather than in frequent ordinary supernovae.

Neutron star mergers, on the other hand, match those requirements. Early theoretical studies had proposed that during the merger of compact objects, tidal forces lead to the

ejection of neutron-rich material [17]. These ejecta were expected to be very neutron-rich and decompress rapidly, ideal for the r-process, enabling the synthesis of heavy neutron-rich nuclei. It was then proposed that the radioactive decay of freshly synthesized nuclei would power a short-lived optical transient. In 1998, Li and Paczyński presented a simple model (LP98) to calculate that possible electromagnetic counterpart. The model predicted a light curve with a bolometric luminosity peak of approximately 10^{44} erg s^{-1} at $t \approx 1$ day after the merger depending on free parameters such as the expansion velocity, the ejected mass and an effective constant for the radioactive heating [18].

1.3 Modern models of kilonovae and observations

The LP98 model assumes an analytical energy generation rate of the form $\dot{\varepsilon}(t) \propto t^{-1}$ and models the electromagnetic emission as a blackbody spectrum. However, these assumptions may not fully capture the physical complexity of the event. As discussed before, the γ-rays produced by nuclear decays can escape without thermalizing, leading to deviations in the predicted emission spectrum. In addition, the actual energy generation rate may deviate from the assumed analytical form, making a more detailed treatment of the spectral evolution of the emission a challenging task.

The first attempt to construct a realistic model of this phenomenon was presented by Metzger et al. [3], who performed self-consistent calculations of the optical and near-infrared counterparts with two major improvements. First, they implemented nuclear reaction network calculations to evolve the abundances and compute the corresponding energy generation rate, $\dot{\varepsilon}(t)$, using both experimental and theoretical reaction and decay rates. Secondly, they performed one-dimensional Monte Carlo radiative transfer simulations, assuming opacities derived from bound–bound transitions of iron (Fe), with corrections applied to the ionization energies, in order to model the resulting light curves. The first results showed an abundance pattern close to the solar system r-process abundances. Also, the energy generation rate was found to follow a power-law dependence, $\dot{\varepsilon}(t) \propto t^{-\alpha}$ with $\alpha \sim 1.1$–1.3, and remarkably robust against variations in the electron fraction ($Y_e \sim 0.05$–0.35), thermodynamic trajectories, and nuclear properties along the r-process path. The second key result demonstrated that the light curves evolve significantly differently from those predicted by the simple blackbody approximation. The simulated light curves exhibited a characteristic "bump-like" behavior on different timescales across various wavelength bands and a peak luminosity three orders of magnitude greater than that of a typical nova, the latter being the reason why the term kilonova was first introduced.

However, radiative transfer simulations face major challenges due to the highly complex opacities arising from the dense line spectra of heavy r-process nuclei. This "opacity problem" has been addressed through several approaches. For instance, in Ref. [19], atomic structure models for representative elements were employed to compute radiative transition rates for tens of millions of lines, enabling more realistic estimates of the opacities for r-process material. A different approach was taken in Ref. [20] where the opacity was estimated by taking into account electron scattering, free–free, bound–free and bound–bound transitions using analytical formulas. These advances led to the identification of distinct kilonova components based on the lanthanide content of the ejecta [21, 22]. Material with a low

electron fraction ($Y_e \leq 0.22$) is expected to produce lanthanide-rich ejecta with high opacity, resulting in a red, longer-lived transient commonly referred to as the red kilonova. In contrast, material ejected with a higher electron fraction ($Y_e \geq 0.3$), typically from the polar regions, evolves into lanthanide-poor ejecta with lower opacity, producing a short-lived, blue optical transient known as the blue kilonova.

Nevertheless, all previous efforts remained purely theoretical and computational, as no observational data were available to validate the models. This situation changed on August 17, 2017, when the Advanced LIGO and Advanced Virgo gravitational-wave detectors made their first observation of a binary neutron star merger (BNSM) [23] with the subsequent identification of the electromagnetic counterparts in the same location, which supported the identification of the event as a BNSM. An approximately 100 s long gravitational-wave signal, GW170817, was followed by a short gamma ray burst (sGRB), GRB 170817A, and an optical transient named SSS17a/AT 2017gfo [24], found in the host galaxy NGC 4993. The source of the signal was detected across the electromagnetic spectrum in the X-ray, ultraviolet, optical, infrared, and radio bands over hours, days, and weeks [25].

The electromagnetic transient SSS17a/AT 2017gfo was of great significance, as it provided the first compelling observational evidence for the nucleosynthesis of heavy nuclei through the r-process. As shown in Ref. [26], the observed emission was well reproduced by a three-component kilonova model: a lanthanide-poor blue component with an opacity of $\kappa = 0.5$ $\mathrm{cm^2\,g^{-1}}$, ejecta mass $M_{\rm ej} \approx 0.020\,M_\odot$ and velocity $v_{\rm ej} = 0.27c$; an intermediate (purple) component with $\kappa = 3$ $\mathrm{cm^2\,g^{-1}}$, $M_{\rm ej} \approx 0.047\,M_\odot$ and $v_{\rm ej} = 0.15c$; and a lanthanide-rich red component with $\kappa = 10$ $\mathrm{cm^2\,g^{-1}}$, $M_{\rm ej} \approx 0.011\,M_\odot$ and $v_{\rm ej} = 0.14c$. Meanwhile, numerous efforts have been made to improve previous kilonova models, motivated by the prospect of obtaining high-resolution data from future observations with the James Webb Space Telescope (JWST). However, performing fully consistent general relativistic simulations that include a realistic treatment of nuclear matter with proper neutrino transport, followed by detailed nucleosynthesis calculations in the ejecta and subsequent radiative transfer simulations to reproduce the observed light curves, remains computationally demanding. Consequently, several practical semi-analytical formulations have been proposed to approximate key physical processes and reduce the computational cost while retaining the essential physics of the problem.

One of the cornerstones of analytical modeling was presented in Ref. [27], which introduced a simplified model not intended to replace full multi-group radiative transfer simulations, but rather to serve as a toy model for qualitatively understanding the evolution of the transient. More refined analytical approaches were proposed shortly thereafter. For instance, Ref. [28] developed a model capable of reproducing multi-band light curves by assuming a time-dependent effective opacity and a uniform power-law energy generation rate. Some models also account for the impact of ejecta morphology on the observed emission, introducing more realistic prescriptions for non-spherical geometries and their influence on the angular dependence of the kilonova light curves [29].

The present panorama of kilonova modeling extends beyond purely analytical formulations and is now better represented by semi-analytical frameworks, such as the one introduced in Ref. [30]. This framework provides a practical approach for generating broadband light curves that account for multiple ejecta components and non-spherical geometries, while incorporating time- and composition-dependent heating rates, thermalization efficiencies, and

opacities.

Such advances in kilonova modeling have opened new possibilities for studying distinct features in the light curves that may contain additional information about the composition of the ejecta. It has been theorized that the late time kilonova emission, occurring weeks to months after the merger, can reveal the radioactive signatures of specific heavy nuclei, rather than the power-law heating that dominates the early time emission [31]. Isotopes such as ^{223}Ra ($t_{1/2} = 11.43$ d) and ^{225}Ac ($t_{1/2} = 10.0$ d), which undergo α-decay, and ^{254}Cf, which undergoes spontaneous fission, have been identified as potential dominant contributors to the late-time heating [32].

These nuclei release large amounts of energy per decay and the α particles thermalize more efficiently than typical β-decay products. Therefore, these nuclei are expected to produce detectable bumps or departures from a smooth power-law decline in the bolometric light curve. However, identifying such "fingerprints" requires sufficiently accurate heating-rate predictions from nucleosynthesis calculations and sensitive, multi-epoch, wide-band observations.

In March 2023, the electromagnetic transient AT 2023vfi, associated with the gamma-ray burst GRB 230307A, was identified as the second confirmed observation of a kilonova [33]. This signal has provided an early opportunity to apply these ideas observationally [34]. JWST follow-up analyses indicate that AT 2023vfi's early evolution resembles AT 2017gfo during the first week, and detailed spectral/photometric modeling has been used to place constraints on the presence and abundances of heavy nuclei in the ejecta. The work presented in Ref. [35] reports upper limits and tentative abundance estimates for isotopes such as ^{222}Rn, ^{223}Ra, ^{224}Ra, and ^{225}Ac. These results illustrate both the promise and the current limitations: while JWST data can probe late-time radioactive contributors, robust identification of specific isotopes requires careful propagation of nuclear-physics (decay-rate and fission-yield) and opacity uncertainties. In this context, one of the goals of this work is to assess whether a more accurate treatment of decay-rate uncertainties can improve the constraints on the abundance of heavy nuclei.

1.4 Objectives and thesis structure

As mentioned before, the main objective of this work is to evaluate the sensitivity of kilonova light curves to the choice of theoretical decay rates prescription. We constructed an ensemble of possible initial thermodynamic trajectories within the ejecta. For each initial condition, we ran six nuclear reaction network calculations with three different β^--decay and two α-decay prescriptions. Then, with the results for the abundances and the heating rate, we modeled the kilonova. We found consistent deviations from the power-law behavior across the ensemble of initial conditions. We also identified the nuclei responsible for such behavior along with their decay chains.

The thesis is divided into three main parts. The introductory part consists of Chapters 1 and 2. Chapter 2 develops the physics needed to understand neutron stars and binary neutron star mergers. Chapters 3, and 4 present the basic theoretical framework from literature for the stages of the merger simulation: the nuclear reaction network evolution and the construction of the kilonova light curve from emission models. The last part, Chapters 5, 6, 7 and 8, will explain the details of how this work constructed the initial thermodynamic trajectories,

the setup we used for the nuclear reaction network calculations with their results[1] and the construction of the kilonova light curves.

[1]Results presented in Chapter 6 have been published in [36].

Chapter 2

From Neutron Stars to Kilonovae

Kilonovae are the main topic of study throughout this thesis. Every chapter and section was built with the objective of developing the intuition on all the processes that take place in kilonovae. However, the necessary astrophysical context will enrich our understanding. Therefore, this chapter will first present the neutron star and its dense matter in the general relativity description. Then the text will move to the problem of finding two neutron stars in a close enough orbit and present a brief discussion of why the existence of such configurations is not trivial. After that, a description of the different phases of the merger will be presented. The different emission mechanisms will be discussed: gravitational waves, short gamma ray bursts and our favorite the kilonova.

2.1 The Neutron Star

Neutron stars contain the densest matter known to exist in a stable state. Their central densities reach several times the nuclear saturation density $n_{\rm sat} \simeq 0.15$ fm^{-3} [37]. In the outer layers of the star, nuclei coexist with a degenerate electron gas. With increasing density, electron captures make the nuclei progressively more neutron-rich until, at the neutron-drip density $n_{\rm drip} \simeq 2.5 \times 10^{-4}$ fm^{-3} ($\rho \simeq 4 \times 10^{11}$ g cm^{-3}), neutrons are no longer bound to the nuclei and form a free neutron gas [38]. At about half the saturation density, the distinction between the inside and the outside of a nucleus effectively vanishes and matter becomes a continuum of nucleons, which marks the transition from crust to core. At the same time, the β equilibrium favors neutrons instead of protons, which leads to a neutron dominated composition.

Observed neutron-star masses range from about 1.1 to slightly above $2\,M_\odot$, and their radii are inferred to lie between roughly 10 and 14 km [39, 40]. However, a free degenerate neutron gas can support stars of at most $\simeq 0.7\,M_\odot$ [41, 42]. To explain the observations accurately, the strongly repulsive short-range part of the nuclear interaction is required. The equation of state (EOS), therefore, has to include nuclear interaction terms, which is a formidable task to calculate.

The microscopic theory of this matter, quantum chromodynamics (QCD), cannot at present provide the EOS in the relevant density range. At the baryon chemical potentials of neutron-star cores, QCD is non-perturbative, and lattice QCD is hampered by the fermion

sign problem at finite baryon density. Controlled calculations exist only in two limits: chiral effective field theory describes nuclear matter up to about 1–2 $n_{\rm sat}$ [43], and perturbative QCD becomes reliable only at densities of order 40 $n_{\rm sat}$, far above those realized in neutron stars [44]. In the intermediate regime, where neutron-star cores reside, the EOS is described by phenomenological models. As a consequence, a first principles description of neutron star cores is still lacking. Matter at several times $n_{\rm sat}$ may consist of nucleons, may contain hyperons, or may undergo a transition to deconfined quark matter, either through a smooth crossover or through a first order hadron-quark phase transition [45].

Let us assume, for instance, that we know the EOS. We can then estimate the radius of the star for a certain range of masses, including GR effects, through the Tolman-Oppenheimer-Volkoff equation:

$$\frac{dP}{dr} = -\frac{G}{r^2}\left(\rho + \frac{P}{c^2}\right)\left(m + 4\pi r^3 \frac{P}{c^2}\right)\left(1 - \frac{2Gm}{c^2 r}\right)^{-1}. \tag{2.1}$$

The equation assumes a static, spherically symmetric perfect fluid. This is a good approximation, but real stars have angular momentum, so rotational corrections will be relevant. Here $\rho(r)$ and $P(r)$ are the density and pressure at a radial distance r. The quantity $m(r)$ is the mass within a sphere of radius r. At this point we have not introduced any EOS yet. The relationship between P and ρ could depend on temperature and composition. Many current EOS models actually present their results for a range of temperatures and electron fractions Y_e. Once the EOS is established, one can find a mass–radius curve. By comparing with observations one can in principle infer constraints on the EOS. Such studies have been done, for example the existence of pulsars with masses of about 2 $M_\odot$ [46] and the radii inferred by NICER [40] constrain the pressure of dense matter up to densities of roughly $2-3\,n_{\rm sat}$ [47]. These measurements, however, constrain the relation between pressure and energy density rather than giving a full description of the matter. For example, hybrid stars (neutron stars with quark cores) can have mass–radius relations nearly indistinguishable from those of purely hadronic stars [48].

2.2 Formation of binary neutron stars

For two neutron stars to merge, three conditions must be met. Both stars must form, which requires two core-collapse supernovae (SNe), and the binary must survive both explosions. Then the two remnants must end up close enough for gravitational-wave (GW) emission to bring them into contact in less than the age of the universe. For typical neutron-star masses, the initial separation must be smaller than about 5 $R_\odot$. This is two to three orders of magnitude smaller than the radii that the progenitor stars reach as supergiants. The central problem of binary-neutron-star formation is therefore how to bring the compact remnants of two massive stars to separations far smaller than the stars themselves once were, without unbinding the system on the way.

In the standard isolated channel, a double neutron star (DNS) descends from a binary of two OB stars that are massive enough to end their lives in a core-collapse SN (see Fig. 2.1). If the stars of the initial binary are too far apart, the resulting DNS will not be close enough to merge. If, on the contrary, the initial stars are close enough, they will interact via stable

or unstable mass transfer. Once one of the stars reaches the end of its life, it expands and transfers part of its mass to its companion. This star then explodes in a SN and leaves a NS, as in Fig. 2.1. If the two stars remain bound after the supernova, they become a high-mass X-ray binary (HMXB). When the second star expands, it engulfs the first star in a phase called common envelope [49]. In that stage, the dynamical friction on the first star as it moves inside the envelope of the second one causes the loss of angular momentum needed to bring the cores closer. This phase is not stable, so eventually the envelope either falls onto the stars or is ejected, leaving behind the first NS and the He core of the second star, which eventually dies in what is called an ultra-stripped SN [50]. Finally, the system results in a DNS close enough to merge. The sequence of events is reviewed in [51].

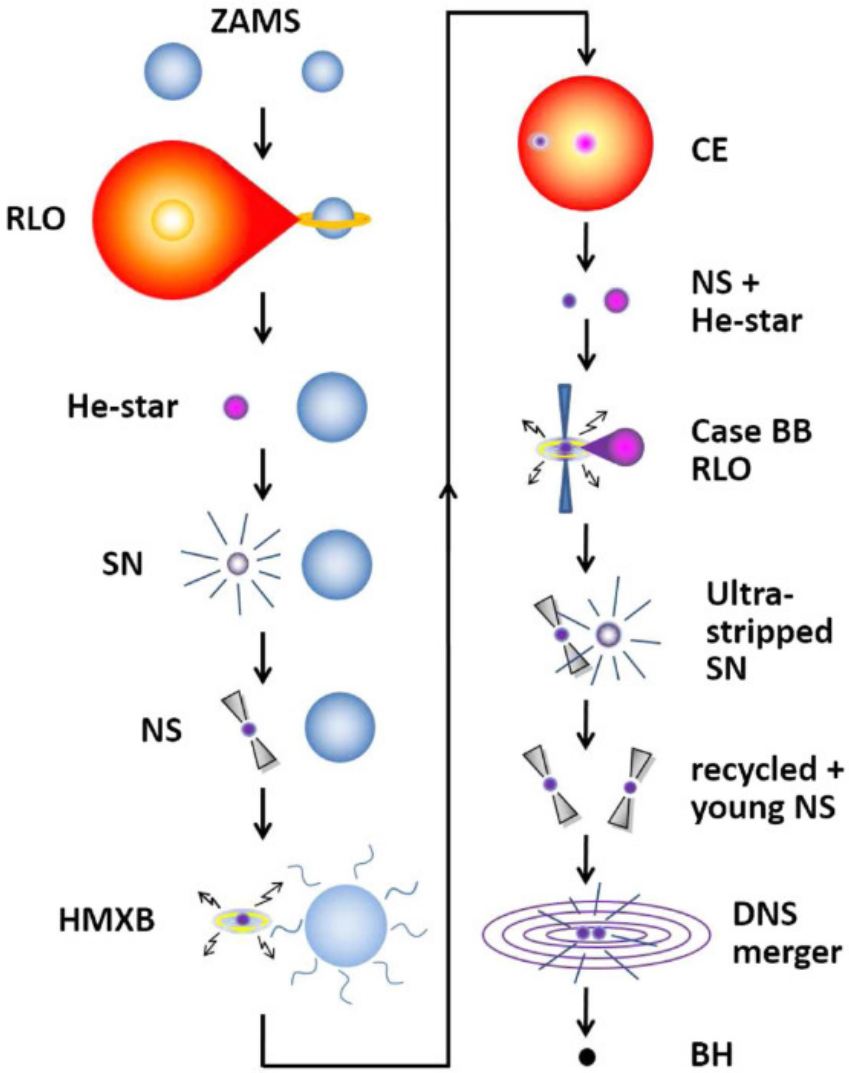


Figure 2.1: Double neutron star (DNS) formation from two OB stars as presented in [51]. ZAMS refers to zero-age main sequence, RLO Roche-lobe overflow, SN supernova, HMXB high-mass X-ray binary and CE common envelope.

2.3 Gravitational-wave-driven inspiral and the maximum separation

Once both neutron stars have formed, the only process that changes the orbit appreciably is the emission of gravitational radiation. Here we will show an estimate of the time to merger at leading (quadrupole) order and the maximum initial separation that allows a merger within the age of the Universe. This calculation is based on the work presented in [52].

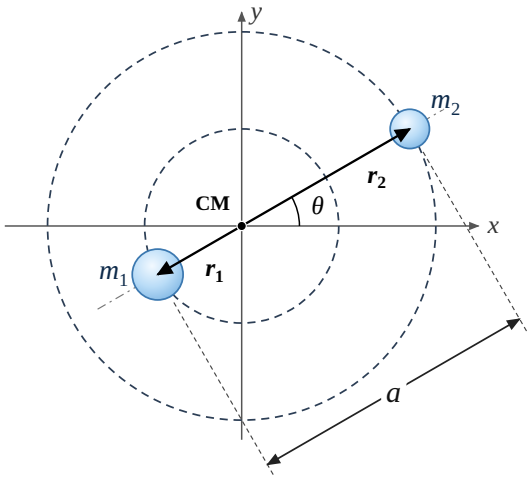


Figure 2.2: Two neutron stars with masses m_1 and m_2 with separation a seen from the center of mass frame.

Consider two point masses m_1 and m_2 on a circular orbit of separation a and angular frequency $\omega = \sqrt{GM/a^3}$, with total mass $M = m_1 + m_2$ and reduced mass $\mu = m_1 m_2/M$. Their position vectors are $\mathbf{r}_1$ and $\mathbf{r}_2$, respectively (see Fig. 2.2). Define the quadrupole mass tensor as

$$Q_{ij} = \sum_l m_l r_{li} r_{lj}, \tag{2.2}$$

summing over the masses m_l with $r_{li} = \mathbf{r}_l \cdot \hat{\imath}$. From this configuration, the positions are

$$\mathbf{r}_1 = -\frac{\mu a}{m_1}\hat{n}\,; \quad \mathbf{r}_2 = \frac{\mu a}{m_2}\hat{n} \quad \text{with} \quad \hat{n} = (\cos(\theta), \sin(\theta), 0). \tag{2.3}$$

So the quadrupole mass tensor components are:

$$Q_{xx} = \frac{\mu^2 a^2}{m_1}\cos^2(\theta) + \frac{\mu^2 a^2}{m_2}\cos^2(\theta) = \mu a^2 \cos^2(\theta) = \frac{\mu a^2}{2}(1 + \cos(2\theta)) \tag{2.4}$$

$$Q_{yy} = \mu a^2 \sin^2(\theta) = \frac{\mu a^2}{2}(1 - \cos(2\theta)) \tag{2.5}$$

$$Q_{xy} = \mu a^2 \sin(\theta)\cos(\theta) = \frac{\mu a^2}{2}(\sin(2\theta)) \tag{2.6}$$

Fortunately, the power radiated in the form of GWs ($L_{\rm GW}$) can be estimated using Eq. (6) of Ref. [52].

$$L_{\rm GW} = \frac{G}{5c^5}\left(\dddot{Q}_{ij}\dddot{Q}_{ij} - \frac{1}{3}\dddot{Q}_{ii}\dddot{Q}_{jj}\right) \tag{2.7}$$

$$L_{\rm GW} = \frac{G}{5c^5}\left(\dddot{Q}_{xx}{}^2 + \dddot{Q}_{yy}{}^2 + 2\dddot{Q}_{xy}{}^2 - \frac{1}{3}(\dddot{Q}_{xx} + \dddot{Q}_{yy})^2\right). \tag{2.8}$$

Since $\dddot{Q}_{xx} = -\dddot{Q}_{yy}$ the last term vanishes and with $\dddot{Q}_{xx}^2 + \dddot{Q}_{yy}^2 + 2\dddot{Q}_{xy}^2 = 32\,\mu^2 a^4\omega^6$ (see Appendix A), the GW luminosity is

$$L_{\rm GW} = \frac{32}{5}\frac{G}{c^5}\mu^2 a^4\omega^6 = \frac{32}{5}\frac{G^4}{c^5}\frac{(m_1 m_2)^2 M}{a^5}. \tag{2.9}$$

The energy radiated is drawn from the orbital energy $E = -Gm_1m_2/(2a)$. Setting $\mathrm{d}E/\mathrm{d}t = -L_{\rm GW}$ gives

$$\frac{\mathrm{d}a}{\mathrm{d}t} = -\frac{\beta}{a^3}, \qquad \beta \equiv \frac{64}{5}\frac{G^3 m_1 m_2 M}{c^5}. \tag{2.10}$$

Equation 2.10 integrates to $a^4(t) = a_0^4 - 4\beta t$, so a binary with initial separation a_0 merges ($a \to 0$) after

$$t_m(a_0) = \frac{a_0^4}{4\beta} = \frac{5}{256}\frac{c^5 a_0^4}{G^3 m_1 m_2 M}. \tag{2.11}$$

The steep a_0^4 dependence is the key point. Halving the separation shortens the inspiral by a factor of sixteen.

Now let us estimate the maximum separation for BNS to merge within the age of the Universe. Note that, since the stars require a prior formation and evolution, this calculation overestimates the maximum distances between stars. Let us require $t_{\rm m} \leq t_{\rm U}$, with $t_{\rm U} \simeq 13.8$ Gyr the age of the Universe

$$a_{\rm max} = (4\beta\, t_{\rm U})^{1/4} = \left(\frac{256}{5}\frac{G^3 m_1 m_2 M}{c^5}\, t_{\rm U}\right)^{1/4}. \tag{2.12}$$

For $m_1 = m_2 = 1.35\,M_\odot$, $m_1 m_2 M = 4.92\,M_\odot^3$ and $\beta = 6.08 \times 10^{27}$ cm^4 s^{-1}. Hence

$$a_{\rm max} \simeq 3.2 \times 10^6 \text{ km} \simeq 4.6\,R_\odot, \tag{2.13}$$

about 0.02 AU. As reference, red supergiants have typical radii of hundreds of $R_\odot$, so having neutron stars closer than $4.6\,R_\odot$ is quite a challenge.

2.4 Mass ejection: tidal, shock-driven, and disk-wind ejecta

At the last instants of the inspiral phase, when the separation becomes comparable to a few stellar radii, the point mass approximation is no longer accurate. From those last instants

to the first ones after the merger the system becomes really complex to study. Physical phenomena and processes such as deformation, dense matter compression and heating, neutrino interactions and the appearance of strong magnetic fields become relevant. Even though current merger simulations include specialized treatment for each of the previously mentioned effects there remain loopholes in our understanding [53, 54]. From the point of view of nucleosynthesis, the important outcome of the merger is the matter that becomes gravitationally unbound. This matter is what we will call the ejecta and is divided into two categories according to the mechanism of ejection [27]. The first is the dynamical ejecta, it is expelled within ~ 10 ms of the merger by tidal torques (tidal ejecta) and by shocks at the collision interface (shock-heated ejecta). The second is the secular or post-merger ejecta, it is expelled after the dynamical ejecta by winds from the remnant and from the accretion disk that surrounds it.

2.4.1 Dynamical ejecta

The study of dynamical ejecta is closely related to numerical relativity simulations. Results from such simulations help us to estimate really important properties of the ejected material such as masses, velocities, distribution, temperatures and electron fraction depending on the sophistication of the simulation. Current results from simulations estimate that the amount of dynamical ejecta lies between $\sim 10^{-4}$ and $\sim 10^{-2} M_{\odot}$, with asymptotic velocities of 0.1–0.3 c [55]. Electron fraction and distribution depend on the exact mechanism. Tidal ejecta usually have low electron fraction ($Y_e < 0.15$) and are ejected mostly through the equator. Shock driven ejecta on the other hand could reach higher electron fractions ($Y_e < 0.4$) and could be emitted at angles close to the poles (see Fig. 2.3). One interesting result is that the channels are not equally likely to dominate. The tidal component dominates for unequal masses and stiff EOS while the shock component dominates for equal masses and soft EOS. In addition, the total ejecta mass correlates with the tidal deformability and with the mass ratio and there are fits calibrated on simulations for such quantities [56].

2.4.2 Post-merger ejecta

Post-merger ejecta is quite interesting since its mechanism relies on the presence of winds from the remnant. So, let us talk a little bit about the remnant. BNSMs have two possible outcomes, they become a black hole or another neutron star. The first outcome happens when there is a collapse and the interesting fact about this is that it could happen at any time during the merger. For example, the binary could become a very heavy unstable neutron star (long-lived remnant) that, after it cools down (typically seconds), collapses to a black hole. The collapse can happen also at the moment of the merger (prompt collapse) or during a viscous phase shortly after (short-lived remnant).

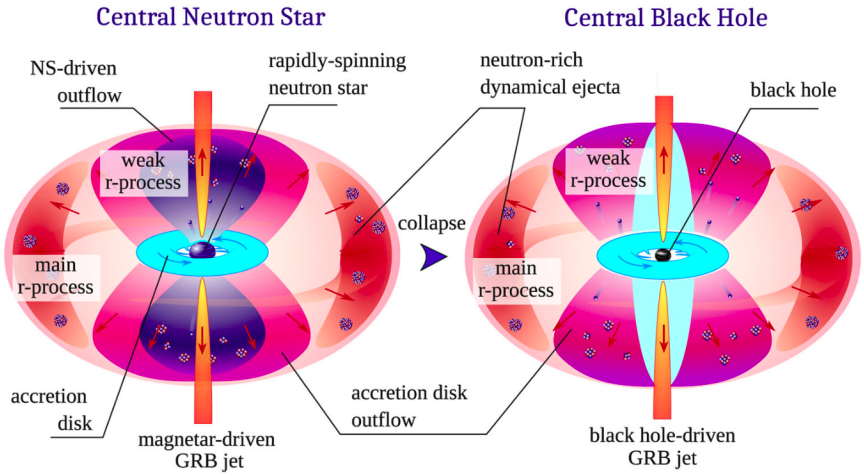


Figure 2.3: Diagram of the remnant from BNSM. Figure taken from Ref. [2].

In all but the most prompt collapses, part of the material with high angular momentum settles into a rotationally supported torus around the central object, with masses $\sim 10^{-3}$–$0.2 M_\odot$. At early times the disk is dense, hot and has low electron fraction. As the disk spreads the viscous heating and the synthesis of α particles and heavier nuclei drive the outflow [57]. As a result part of the disk accumulates enough kinetic energy to unbind from the remnant contributing to the kilonova.

2.5 Weak interactions and the electron fraction of the ejecta

Before the merger, the stars can be described as cold neutron rich matter. At the merger, as we mentioned, the matter heats up and several effects take place. As the matter moves from the β equilibrium, neutrinos become important since they control the Y_e distribution across the ejecta. This is why neutrino transport schemes are important in merger simulations. Simulations without them mispredict the Y_e and are not useful for abundance calculations. This section will establish first why Y_e is important and then how neutrinos control it.

The outcome of the r-process in an expanding fluid element is set mainly by three quantities evaluated when nuclear statistical equilibrium freezes out: the electron fraction Y_e, the entropy per baryon s, and the expansion timescale $\tau_{\rm exp}$ [22]. They fix the neutron-to-seed ratio at the onset of neutron captures. Merger ejecta have moderate entropies, $s \sim 10$–$30\, k_B$ per baryon, and short expansion times, $\tau_{\rm exp} \sim$ ms, so Y_e is the dominant parameter. By

charge neutrality,

$$Y_e = \frac{n_{e^-} - n_{e^+}}{n_b} = \sum_i Z_i Y_i, \qquad Y_i = \frac{X_i}{A_i}, \tag{2.14}$$

where X_i is the mass fraction (ρ_i/ρ), A_i and Z_i are the mass number and proton number of species i respectively and $n_{e^\pm}, n_b$ are number densities of positrons, electrons and baryons. Strong and electromagnetic reactions conserve Y_e. It changes only through charged-current weak interactions.

In a cold neutron star the neutrinos escape freely ($\mu_\nu = 0$) and matter is in neutrinoless β equilibrium, $\mu_n = \mu_p + \mu_e$[1]. Because μ_e grows rapidly with density, equilibrium favors neutrons. In the inner crust and outer core, from which the dynamical ejecta originate, Y_e has values just above neutron drip to ≈ 0.03–0.1 at $\rho \sim 10^{13}$–10^{14} g cm^{-3}. If this matter were ejected without further weak processing, it would have a neutron-to-seed ratio of several hundred and produce a robust abundance pattern from the second peak up to the actinides [58]. During and after the merger, however, the ejecta are heated and irradiated, and two classes of reactions can change Y_e:

$$e^- + p \rightleftarrows n + \nu_e, \qquad e^+ + n \rightleftarrows p + \bar{\nu}_e, \tag{2.15}$$

$$\nu_e + n \to p + e^-, \qquad \bar{\nu}_e + p \to n + e^+. \tag{2.16}$$

Captures on heavy nuclei and the inverse reactions with thermal neutrinos also contribute in dense, opaque regions, but for the ejecta the reactions on free nucleons dominate.

For matter composed of free nucleons the abundances are $Y_p = Y_e$ and $Y_n = 1 - Y_e$, and the reactions 2.15–2.16 give

$$\frac{dY_e}{dt} = \underbrace{(\lambda_{\nu_e n} + \lambda_{e^+ n})}_{\equiv \lambda_+}(1 - Y_e) - \underbrace{(\lambda_{\bar{\nu}_e p} + \lambda_{e^- p})}_{\equiv \lambda_-} Y_e. \tag{2.17}$$

Where λ_{ab} is the rate of the reaction reactants ab. For slowly varying rates, Y_e relaxes exponentially to the instantaneous equilibrium value

$$Y_e^{\rm eq} = \frac{\lambda_+}{\lambda_+ + \lambda_-} \quad \text{on the timescale} \quad \tau_{\rm weak} = \frac{1}{\lambda_+ + \lambda_-}. \tag{2.18}$$

Whether the ejecta reach equilibrium is decided by the ratio $\tau_{\rm exp}/\tau_{\rm weak}$. If $\tau_{\rm exp} \gg \tau_{\rm weak}$, Y_e follows $Y_e^{\rm eq}$; if $\tau_{\rm exp} \ll \tau_{\rm weak}$, Y_e is frozen. Because both the temperature and the neutrino flux fall quickly as the matter expands, every fluid element eventually freezes out. This is why the slow expanding post-merger ejecta reach higher Y_e than the rapid expanding tidal ejecta which is cold and fast and keeps Y_e similar to the interior of the star before the merger.

2.6 Electromagnetic counterparts

GW170817 showed that a BNS merger is accompanied by emission across the whole electromagnetic spectrum [59]. The gamma-ray burst GRB 170817A was detected 1.7 s after

[1] Here μ refers to chemical potentials

the merger [60]. The optical counterpart AT2017gfo was found 10.9 h later in the galaxy NGC 4993 at $\simeq 40$ Mpc [24]. In addition, an X-ray and radio afterglow appeared days later and kept brightening for several months. At this point we have briefly discussed the formation of the binary, the GW emission, the mass ejection channels and the initial conditions of the ejected material. Now it is time to talk about electromagnetic emissions.

After the merger (seconds) the first emission generated is the gamma ray burst from a relativistic jet, which corresponds to a non thermal radiation. As the density decreases and the ejecta approaches an optically thin regime, thermalized photons escape and build the kilonova (days). Also, some non thermal afterglows are produced when the jet and the ejecta decelerate (hours).

2.6.1 GRBs

Gamma ray bursts (GRBs) are high energy photons released from the polar regions by ultra-relativistic, collimated outflows called jets launched by the remnant. This could be an accreting black hole, which extracts its energy through magnetic fields [61], or a rapidly rotating magnetized neutron star. The photons are produced from the acceleration of the charges inside the jet. When the jet drives into the surrounding medium it decelerates and produces a non thermal afterglow.

However, GRBs are not unique to mergers. Some GRBs are associated with the deaths of massive stars. In the collapsar model the core of the rapidly rotating, stripped massive star collapses to a black hole surrounded by a massive accretion disk, which powers a jet for tens of seconds [62, 63]. The working assumption was therefore that the origin of the GRBs can be identified by the duration. Short GRBs come from mergers and long GRBs from collapsars. But in recent years this was proved wrong. GRB 230307A had a prompt phase of tens of seconds followed by a red kilonova [5, 33], and GRB 200826A had a duration of 0.65s and was followed by a supernova [64]. The duration of a burst is therefore not a reliable indicator of its progenitor. Now the central open question is how a merger can power a jet for tens of seconds, when the accretion time of a typical post-merger disk is $\lessapprox 1$ s.

2.6.2 Kilonovae

Kilonovae belong to the thermal radiation emission since their origin comes from the thermalization of the energy from radioactive decays. However its emission is not exactly the one of a black body, it could have deviations, so a kilonova is a quasi-thermal transient. Most kilonova models, from analytic like LP98 to multidimensional radiative transfer, are built from the same few ingredients: homologous expansion of the ejecta, a radioactive heating rate and its thermalization efficiency, and opacity set by the composition and photon diffusion out of an expanding medium.

Homologous expansion

At early stages the dynamical ejecta is accelerated but shortly after the pressure forces become negligible and every fluid element moves ballistically, $r = vt$. One model for the

density that encapsulates this behavior is the one presented in Ref. [22]

$$\rho(\tau_{\rm exp}, t) = \left\{ \begin{array}{ll} \rho_0 e^{-t/\tau_{\rm exp}} & \text{if } t \leq 3\tau_{\rm exp} \\ \rho_0 \left(\frac{3\tau_{\rm exp}}{et}\right)^3 & \text{if } t \geq 3\tau_{\rm exp} \end{array} \right\}. \tag{2.19}$$

where ρ_0 is an initial density and $\tau_{\rm exp}$ the expansion time. Notice that this model represents a smooth transition from an exponential expansion to a ballistic expansion.

Radioactive heating

As nuclei decay part of their energy stored as mass is released in the form of kinetic energy and photons. Part of the kinetic energy escapes from the system in form of neutrinos and the rest thermalizes through plasma interactions. So effectively not all the energy from decays is used to power the transient. This subsection focuses on the energy and the next subsection will talk about thermalization. Let us first define the nuclear energy released per unit mass from decays, $\dot{\epsilon}$, as

$$\dot{\epsilon}(t) = \sum_{i,x} \frac{Q_{i,x}}{M_{ej}} \lambda_{i,x} N_i(t) = \sum_{i,x} \frac{Q_{i,x}}{m_i} \lambda_{i,x} X_i(t) \tag{2.20}$$

where the index i represents the sum over all the nuclear species and the index x represents the sum over decay modes. $Q_{i,x}$ is the energy released by species i that decay through the decay mode x. $\lambda_{i,x}$ is the decay constant through the decay mode x of species i. X_i is the mass fraction and m_i is the mass of a nucleus i. The alternative version uses M_{ej} (the mass of the ejecta) and N_i the total number of nuclei of species i. At this point this equation does not account for thermalization effects. This definition is completely consistent with the standard definition of heat per baryon due to changes in the composition.

$$\dot{q} = -\sum_i (\mu_i + Z_i \mu_e) \dot{Y}_i \tag{2.21}$$

Where Y_i is the number fraction defined as n_i/n which is the number density of i over the number density of baryons, μ_i is the chemical potential of species i. Notice that Eq. 2.20 only accounts for decays while the Eq. 2.21 accounts for any changes in the compositions (decay and reactions). They are consistent in the sense that if we restrict Eq. 2.21 to account for changes only through decays, that means establish an expression for $\dot{Y}_i$, we get Eq. 2.20. Now the issue becomes how to correctly estimate the abundances (X_i,Y_i) as a function of time while taking into account all reactions and decays involved. That problem is solved by implementing a nuclear reaction network which we will treat in detail in Chapter 3. So kilonova models usually implement a reaction network or at least some fits like power laws.

The power-law behavior of $\dot{q}$ can be understood with a statistical argument [3, 65]. Due to the large number of species Eq. 2.20 can be approximated as an integral. From Fermi's theory of allowed β decay the rate scales as, $\lambda \propto Q^5$. Also, $Q \propto \eta$ with $\eta = N - Z$ (neutron excess). Assuming that number of nuclei per neutron excess interval is constant $dN \propto d\eta$[2] it is easy to see that

$$dN = N_0 d\eta = N_0 \lambda^{-4/5} d\lambda, \tag{2.22}$$

[2]The N presented here represents number of nuclei not neutrons

where N_0 is a normalization constant. Let us now calculate the heating rate as follows:

$$\dot{\epsilon} = \sum_i \frac{Q_i \lambda_i}{M_{ej}} N_i(t) \approx \int_0^\infty \frac{1}{M_{ej}} \lambda^{1/5} \lambda N_0 e^{-\lambda t} \lambda^{-4/5} d\lambda \tag{2.23}$$

$$\dot{\epsilon} \propto t^{-7/5} \tag{2.24}$$

in agreement with the analysis of Hotokezaka et al. [65]. Network calculations confirm this behavior, $\dot{\epsilon} \simeq 10^{10}\ \mathrm{erg\,g^{-1}\,s^{-1}}\ (t/\mathrm{day})^{-1.3}$ [3, 58]. A widely used fit is [58]

$$\dot{\epsilon}(t) = \dot{\epsilon}_0 \left[\frac{1}{2} - \frac{1}{\pi} \arctan\left(\frac{t - t_0}{\sigma}\right)\right]^\alpha, \tag{2.25}$$

$$\dot{\epsilon}_0 = 2 \times 10^{18}\ \mathrm{erg\,g^{-1}\,s^{-1}},\ t_0 = 1.3\ \mathrm{s},\ \sigma = 0.11\ \mathrm{s},\ \alpha = 1.3, \tag{2.26}$$

which reproduces the plateau during the r-process and the subsequent power law.

Thermalization

Only the fraction of the decay energy that is deposited in the ejecta can be radiated. The effective heating rate is $\dot{\varepsilon}(t) = \sum_x f_\mathrm{x}(t)\, \dot{\epsilon}_x(t)$, with $\dot{\epsilon}_x(t) = \sum_i \dot{\epsilon}_{i,x}(t)$ and f_x the thermalization function of decay mode x. Since thermalization depends on the nature of the decay products then $f_x(t) = \sum_p \chi_p\, f_p(t)$ where χ_p is the average fraction of the energy carried by the product p and p can be γ, α, β or fission daughters. The f_p function is the specific thermalization for the product p. Gamma rays deposit their energy if the ejecta are optically thick to them. A convenient interpolation is $f_\gamma \simeq 1 - \exp(-t_\gamma^2/t^2)$ where t_γ is a function of the mass, velocity and opacity of the ejecta [4, 66]. Most of the gamma-ray energy therefore escapes after about a day, around the time of the peak.

Electrons, α particles and fission fragments are confined by magnetic fields and lose energy by ionization and excitation, at a rate proportional to the density. A convenient way to approximate the thermalization is presented in Ref. [67],

$$f_e(t) \simeq \left(1 + \frac{t}{t_e}\right)^{-n}, \qquad n \simeq 1\text{–}1.5. \tag{2.27}$$

Here t_e is a timescale that depends on the dynamics and mass of the ejecta. This formula can be applied to any charged particle by using different values for n.

Opacity

The free electrons provide only a small opacity, because the ejecta are cool and only weakly ionized. The opacity of kilonova ejecta is instead dominated by the very large number of bound–bound transitions of the r-process elements which are largely unknown. The first models and radiative transfer simulations used relatively low opacities assuming that the optical properties of the ejecta matched the ones from ordinary SNe, mostly related to iron group lines. Such assumptions were later proved wrong as observations matched a kilonova with a high opacity component. Now, one may think that opacity is then related to the number of electrons an atom has but it is rather the large number of distinct ways of moving

electrons within the open shells that increases the opacity. As reference, the number of ways of placing n electrons in a shell with $g = 2(2\ell+1)$ states is $\binom{g}{n}$, at most 252 for a d shell ($g = 10$) but 3432 for an f shell ($g = 14$). The lanthanides and actinides, with open $4f$ and $5f$ shells, therefore have vastly more levels and lines than the iron group, and an opacity higher by one to two orders of magnitude, concentrated at optical and ultraviolet wavelengths [19, 68]. In grey models this is captured by $\kappa \simeq 0.5$–$1\ \mathrm{cm^2\,g^{-1}}$ for lanthanide-poor ejecta ($Y_e \gtrsim 0.3$) and $\kappa \simeq 10$–$30\ \mathrm{cm^2\,g^{-1}}$ for lanthanide-rich ejecta ($X_{\mathrm{La}} \gtrsim 10^{-2}$, $Y_e \lesssim 0.25$)[3].

Diffusion and the time of the peak

Photons produced in the ejecta escape by diffusion. For a uniform sphere the optical depth[4] is $\tau = \kappa\rho R = 3\kappa M/(4\pi v^2 t^2)$, and the diffusion time (the time needed for the photons to reach the surface) is $t_d \simeq \tau R/c$,

$$t_d = \frac{3\kappa M_{ej}}{4\pi\, c\, v\, t} \equiv \frac{\tau_m^2}{t}, \qquad \tau_m \equiv \left(\frac{3\kappa M_{ej}}{4\pi\, c\, v}\right)^{1/2}. \tag{2.28}$$

At early times $t_d \gg t$ and the radiation is trapped and degraded by adiabatic expansion. The emission peaks roughly when $t \simeq \tau_m$.

At this point ends our introductory stage. We have studied what a neutron star is, the different stages of a merger, their mass ejection and we have discussed briefly the GRBs and the kilonovae. The next chapter is going to be fully dedicated to understand the physics involved when implementing a nuclear reaction network. We will cover from nuclear statistical equilibrium to abundance evolution employing decay rates and discuss the thermodynamics involved in the evolution. After that we will retake the discussion on kilonovae and explain in detail the model we will employ in this work. In Chapter 5 we will make use of the theoretical framework built and present how we use it in our research.

[3] X_{La} is defined as the sum of the mass fractions of all lanthanides.

[4] This τ is different from the expansion time τ_{exp} and here we are assuming that the ejecta is already in the ballistic regime (expanding with constant velocity).

Chapter 3

r-Process Nucleosynthesis

In Chapter 2 we characterized the ejecta of a merger by three quantities, the electron fraction Y_e, the entropy per baryon s and the expansion timescale $\tau_{\rm exp}$, and we saw that the kilonova is powered by the radioactive decay of the nuclei synthesized in them. This chapter connects the two. Its purpose is to present the physics contained in a nuclear reaction network such as WinNet [69], and to show how the nuclear inputs of the network determine the composition and the radioactive heating of the ejecta. Section 3.1 formulates the network equation and the general relations that its reaction rates satisfy. Section 3.2 studies the high-temperature limit of the network, nuclear statistical equilibrium, and its breakdown as the ejecta expand. Section 3.3 describes the r-process itself: the neutron-to-seed ratio, the reaction path and fission cycling. Finally, Sec. 3.4 describes the radioactive energy release and its feedback on the temperature of the ejecta. Extensive reviews of the subject can be found in Refs. [1, 70].

3.1 Abundances, reaction rates and the network equation

A nuclear reaction network follows the composition (species of nuclei, electrons, positrons etc.) of a fluid element as it evolves through a set of reactions and decays. In this section we define the variables that describe the composition, the rates of the different types of reactions and the system of equations that governs the composition. We also present the detailed balance, which fixes every reverse reaction in terms of its forward reaction. This section follows Refs. [69, 71, 72].

3.1.1 Abundances

Consider a fluid element with baryon number density n_b that contains nuclear species i, with Z_i protons, N_i neutrons and mass number $A_i = Z_i + N_i$, at number density n_i. Free neutrons and protons are included as the species $(Z, A) = (0, 1)$ and $(1, 1)$. As in Chapter 2, the abundance of species i is its number per baryon,

$$Y_i = \frac{n_i}{n_b} = \frac{n_i}{\rho N_A}, \tag{3.1}$$

where $\rho \equiv m_u n_b$ is the baryon mass density and m_u the atomic mass unit. The second equality uses $N_A m_u = 1\,\mathrm{g\,mol^{-1}}$. Notice that it is not an exact equation but the difference is

of the order of $\sim 10^{-10}$. The mass fraction is $X_i = (m_i/m_u)\, Y_i = (m_i n_i / m_u n_b) = \rho_i/\rho$. Since $n_b = \sum_i A_i n_i$ and charge neutrality relates the net number of electrons to the number of protons, the abundances satisfy

$$\sum_i A_i Y_i = 1, \qquad \sum_i Z_i Y_i = Y_e, \tag{3.2}$$

the second relation being the definition of Y_e given in Chapter 2. The advantage of Y_i over n_i is that it is unaffected by the expansion of the fluid; as shown below, only nuclear reactions change it.

3.1.2 Reaction rates

The rate r of a reaction is the number of reactions per unit volume and unit time. Reactions are classified by the number of nuclei in the entrance channel.

One-body reactions. Decays, photodisintegrations, lepton captures[1] and neutrino absorptions involve a single nucleus j and have rates $r_j = \lambda_j n_j$. For a radioactive decay, $\lambda_j = \ln 2 / T_{1/2}$ with $T_{1/2}$ the half-life, literature usually assumes a half-life independent of the environment but studies show that half-lives for specific decay modes depend on the environment [73]. In a photodisintegration the nucleus absorbs a photon from the thermal radiation field, so that

$$\lambda_{j,\gamma}(T) = \frac{1}{\pi^2 (\hbar c)^3} \int_0^\infty \frac{\sigma_{j,\gamma}(E_\gamma)\, c\, E_\gamma^2}{\exp\left(E_\gamma / k_B T\right) - 1}\, \mathrm{d}E_\gamma \tag{3.3}$$

depends only on the temperature [2]. Electron and positron captures depend on T and on the electron density ρY_e, and neutrino absorptions on the neutrino number flux F_ν through $\lambda_\nu = \langle \sigma_\nu \rangle F_\nu$, with $\langle \sigma_\nu \rangle$ the cross section averaged over the neutrino spectrum. Together with β decays, these are the reactions that change Y_e in Eq. 2.17.

Two-body reactions. For two nuclei j and k the rate is

$$r_{j,k} = \frac{n_j n_k}{1 + \delta_{jk}} \langle \sigma v \rangle_{j,k}, \tag{3.4}$$

where the Kronecker delta avoids counting pairs of identical particles twice. For non-degenerate nuclei the velocities follow Maxwell–Boltzmann distributions, and the average of σv reduces to an average over the centre-of-mass energy E,

$$\langle \sigma v \rangle_{j,k} = \left(\frac{8}{\pi m_{jk}} \right)^{1/2} \frac{1}{(k_B T)^{3/2}} \int_0^\infty \sigma_{j,k}(E)\, E\, e^{-E/k_B T}\, \mathrm{d}E, \tag{3.5}$$

with m_{jk} the reduced mass. Two limits are relevant in this work. For neutron captures at low energy the s-wave cross section follows the $1/v$ law, so that σv, and hence $\langle \sigma v \rangle$, is nearly

[1] The name one-body reaction only refers to the number of nuclei, electron captures still count as one-body.
[2] Here we assume that photons are thermalized so they obey a Planck distribution.

independent of temperature. For charged particles the cross section contains the Coulomb penetrability $\exp(-2\pi\eta)$, with $\eta = Z_j Z_k e^2/(\hbar v)$ the Sommerfeld parameter, and the rate falls steeply as the temperature drops. This difference is the reason why, in expanding ejecta, charged-particle reactions stop long before neutron captures do (Sec. 3.2).

Three-body reactions. Reactions such as the triple-α process, $3\,{}^4\mathrm{He} \to {}^{12}\mathrm{C}$, or $\alpha\alpha n \to {}^9\mathrm{Be}$ proceed through a short-lived intermediate state (${}^8\mathrm{Be}$) and are treated as effective three-body reactions with rate $r_{j,k,l} = n_j n_k n_l \langle jkl\rangle/(1+\Delta_{jkl})$, where $1+\Delta_{jkl}$ counts the permutations of identical particles, e.g. $1+\Delta = 3!$ for the triple-α reaction [69].

Stellar rates and partition functions. In a hot plasma the nuclei are not only in their ground states. Excited states are thermally populated, and the cross sections in Eqs. 3.3 and 3.5 are averages over this population, such rates are called stellar rates [74, 75]. The population is described by the partition function

$$G_i(T) = \sum_l \frac{2J_l + 1}{2J_0 + 1} \exp\left(-\frac{E_l}{k_B T}\right), \tag{3.6}$$

where the sum runs over the states l of nucleus i with spin J_l and excitation energy E_l, and J_0 is the spin of the ground state.

Parametrization. Rates computed from Eqs. 3.3 and 3.5, either from experimental data or from theory, are distributed in fitted form. The most widely used format, that of the JINA REACLIB library [76], writes each rate as

$$\lambda = \exp\left[a_0 + \sum_{n=1}^{5} a_n T_9^{(2n-5)/3} + a_6 \ln T_9\right], \tag{3.7}$$

where $T_9 = T/10^9\,\mathrm{K}$ and λ stands for λ, $N_A\langle\sigma v\rangle$ or $N_A^2\langle jkl\rangle$ for one-, two- and three-body reactions, respectively.

Electron screening. Free electrons partially screen the Coulomb repulsion between two nuclei. So we need to include a factor to account for that effect. The strength of the effect is measured by the ion-coupling parameter [69]

$$\Gamma_{12} = \frac{2Z_1 Z_2}{Z_1^{1/3} + Z_2^{1/3}} \frac{e^2}{k_B T} \left(\frac{4\pi n_e}{3}\right)^{1/3} \simeq 4.55 \times 10^{-4} \frac{Z_1 Z_2}{Z_1^{1/3} + Z_2^{1/3}} \frac{(\rho Y_e)^{1/3}}{T_9}, \tag{3.8}$$

with $n_e = \rho N_A Y_e$ and ρ in $\mathrm{g\,cm^{-3}}$. Under the conditions in which charged-particle reactions are active in merger ejecta ($T_9 \gtrsim 3$, $\rho \lesssim 10^8\,\mathrm{g\,cm^{-3}}$) one has $\Gamma_{12} \lesssim 1$, and screening is a modest correction. It does not affect neutron-induced reactions, which dominate the r-process.

3.1.3 The network equation

Let C_i be the net number of nuclei i created per unit volume and unit time by all reactions,

$$C_i = \sum_j N^i_j \, \lambda_j n_j + \sum_{j,k} N^i_{j,k} \, \frac{n_j n_k \langle \sigma v \rangle_{j,k}}{1+\delta_{jk}} + \sum_{j,k,l} N^i_{j,k,l} \, \frac{n_j n_k n_l \langle jkl \rangle}{1+\Delta_{jkl}}, \tag{3.9}$$

where the sums run over all one-, two- and three-body reactions (reactions with the same reactants but different products are counted separately), and the integers N^i_j, $N^i_{j,k}$ and $N^i_{j,k,l}$ give the number of nuclei i produced (positive) or destroyed (negative) in each reaction. For example, in the capture ${}^A Z(n,\gamma)^{A+1}Z$ one has $N = -1$ for the neutron and for the target, and $N = +1$ for the product. For reactions with many possible products, such as fission or β-delayed neutron emission, N^i is replaced by the average number of nuclei i produced per reaction, i.e. by the fragment yield or by the emission probability.

In the absence of diffusion all species move with the fluid velocity $\mathbf{v}$, and

$$\frac{\partial n_i}{\partial t} + \nabla \cdot (n_i \mathbf{v}) = C_i, \qquad \frac{\partial n_b}{\partial t} + \nabla \cdot (n_b \mathbf{v}) = 0, \tag{3.10}$$

the second equation being baryon number conservation. Writing $n_i = Y_i n_b$ in the first equation and subtracting Y_i times the second one gives

$$n_b \left(\frac{\partial Y_i}{\partial t} + \mathbf{v} \cdot \nabla Y_i \right) \equiv n_b \frac{\mathrm{d}Y_i}{\mathrm{d}t} = C_i, \tag{3.11}$$

where $\mathrm{d}/\mathrm{d}t$ is the Lagrangian derivative along the fluid element. The terms proportional to $\nabla \cdot \mathbf{v}$ cancel identically, which confirms that Y_i changes only through reactions.[3] Inserting Eq. 3.9 with $n_j = \rho N_A Y_j$ into Eq. 3.11 yields the network equation [69]

$$\frac{\mathrm{d}Y_i}{\mathrm{d}t} = \sum_j N^i_j \, \lambda_j Y_j + \sum_{j,k} \frac{N^i_{j,k}}{1+\delta_{jk}} \, \rho N_A \langle \sigma v \rangle_{j,k} \, Y_j Y_k + \sum_{j,k,l} \frac{N^i_{j,k,l}}{1+\Delta_{jkl}} \, \rho^2 N_A^2 \langle jkl \rangle \, Y_j Y_k Y_l, \tag{3.12}$$

one equation for each species. The density enters explicitly through the factors ρN_A, and the temperature through the rates.

Equation 3.12 preserves the constraints of Eq. 3.2. Multiplying it by A_i and summing over i, the contribution of each reaction is proportional to $\sum_i A_i N^i$, the net change of baryon number in that reaction, which vanishes; hence $\mathrm{d}(\sum_i A_i Y_i)/\mathrm{d}t = 0$. Multiplying by Z_i instead, strong and electromagnetic reactions give $\sum_i Z_i N^i = 0$, whereas a weak reaction changes the total nuclear charge by $\Delta Z = \pm 1$, because leptons are not part of the set $\{Y_i\}$. Therefore

$$\frac{\mathrm{d}Y_e}{\mathrm{d}t} = \sum_{w \,\in\, \text{weak}} \Delta Z_w \, \frac{r_w}{n_b}, \tag{3.13}$$

where the sum runs over all weak reactions w with rates r_w. Restricted to free nucleons, Eq. 3.13 is Eq. 2.17 of Chapter 2.

[3] The same argument holds in general relativity. With the four-velocity u^μ of the fluid, $\nabla_\mu (n_i u^\mu) = C_i$ and $\nabla_\mu (n_b u^\mu) = 0$ give $u^\mu \nabla_\mu Y_i = \mathrm{d}Y_i/\mathrm{d}\tau = C_i/n_b$, where τ is the proper time of the fluid element and n_i, n_b and C_i are evaluated in its rest frame.

Equation 3.12 describes a single fluid element that does not exchange matter with its neighbours. It must be supplemented with the history of that element, $\rho(t)$ and $T(t)$, and, if neutrino reactions are included, with the neutrino fluxes. This history is either recorded by tracer particles in a hydrodynamic simulation or given by a parametrized expansion such as Eq. 2.19. The temperature is not entirely external, however, since the energy released by the reactions heats the matter. This self-heating is going to be very important and is discussed in Sec. 3.4. For r-process calculations the network contains ~ 6800 nuclei, from free nucleons up to $Z \approx 112$, connected by ~ 72000 reactions [69]. The system is non-linear, because the two-body terms are quadratic in the abundances, and very stiff: its rates range from $\lambda \gtrsim 10^{10}\,\mathrm{s}^{-1}$ for photodisintegrations at a few GK to $\lambda \approx 5 \times 10^{-18}\,\mathrm{s}^{-1}$ for ^{238}U. It is therefore integrated with implicit methods, for which we refer to Refs. [69, 71, 72].

3.1.4 Detailed balance

Every reaction has a reverse, and the two rates are not independent. Consider the radiative capture $a + b \to c + \gamma$, with Q-value $Q = (m_a + m_b - m_c)c^2 > 0$, and its reverse, the photodisintegration $c + \gamma \to a + b$. In complete thermal equilibrium the two rates balance, $\lambda_{c,\gamma} n_c = n_a n_b \langle \sigma v \rangle_{a,b} / (1 + \delta_{ab})$, so that

$$\lambda_{c,\gamma} = \frac{1}{1+\delta_{ab}} \left(\frac{n_a n_b}{n_c} \right)_{\mathrm{eq}} \langle \sigma v \rangle_{a,b}. \tag{3.14}$$

The equilibrium ratio follows from the number density of a non-degenerate species with internal degeneracy $g_i G_i(T)$,

$$n_i = g_i G_i(T) \left(\frac{m_i k_B T}{2\pi\hbar^2} \right)^{3/2} \exp\left(\frac{\mu_i - m_i c^2}{k_B T} \right), \tag{3.15}$$

where $g_i = 2J_0 + 1$ and μ_i is the chemical potential including the rest mass, together with the equilibrium condition $\mu_c = \mu_a + \mu_b$ (photons have $\mu_\gamma = 0$). Inserting Eq. 3.15 into Eq. 3.14, the chemical potentials cancel and

$$\lambda_{c,\gamma} = \frac{g_a g_b}{(1+\delta_{ab})\, g_c} \frac{G_a(T)\, G_b(T)}{G_c(T)} \left(\frac{m_a m_b k_B T}{m_c 2\pi\hbar^2} \right)^{3/2} \exp\left(-\frac{Q}{k_B T} \right) \langle \sigma v \rangle_{a,b}. \tag{3.16}$$

The same argument applied to a reaction $a + b \to c + d$ gives

$$\langle \sigma v \rangle_{c,d} = \frac{1+\delta_{cd}}{1+\delta_{ab}} \frac{g_a g_b\, G_a G_b}{g_c g_d\, G_c G_d} \left(\frac{m_a m_b}{m_c m_d} \right)^{3/2} \exp\left(-\frac{Q}{k_B T} \right) \langle \sigma v \rangle_{a,b}, \tag{3.17}$$

without the factor $T^{3/2}$, because the number of particles is the same on both sides.

Although Eqs. 3.16 and 3.17 were derived in equilibrium, they hold at all times, since rate coefficients depend only on the temperature and on the cross sections, not on the abundances. Microscopically they follow from the reciprocity theorem between the cross sections connecting individual nuclear states; the partition functions appear because both rates are averages over thermally populated states [74, 75]. Detailed balance therefore holds for stellar rates, not for rates computed for nuclei in their ground states.

Equation 3.16 has two consequences. First, the reverse rate depends exponentially on the Q-value, therefore on the masses. The Q-value of a neutron capture is the neutron separation energy of the product, $Q = S_n(Z, A+1)$. An error of only 0.5 MeV in S_n changes $\lambda_{\gamma n}$ at $T_9 = 1$ by a factor $e^{5.8} \approx 330$. This exponential sensitivity determines the location of the r-process path (Sec. 3.3) and makes masses an important nuclear input far from stability. Second, the network is consistent with equilibrium only if the forward rates, the reverse rates and the equilibrium composition of Sec. 3.2 are computed with the same masses.

3.1.5 The network during the r-process

During the r-process the relevant reactions are neutron captures and their inverse photodisintegrations, β^- decays including β-delayed neutron emission, α decays and fission. Charged-particle reactions have already frozen out (Sec. 3.2). Writing $Y(Z, A)$ for the abundance of the nucleus with Z protons and mass number A, Eq. 3.12 takes the form

$$\begin{aligned}\frac{\mathrm{d}Y(Z,A)}{\mathrm{d}t} = &\underbrace{\lambda_{n\gamma}(Z,A-1)\,Y(Z,A-1)}_{\text{neutron capture}} + \underbrace{\lambda_{\gamma n}(Z,A+1)\,Y(Z,A+1)}_{\text{neutron emission}} \\ &+ \underbrace{\sum_{k\geq 0} P_{kn}(Z-1,A+k)\,\lambda_\beta(Z-1,A+k)\,Y(Z-1,A+k)}_{\beta^-\text{-delayed k-neutrons emission}} \\ &+ \underbrace{\lambda_\alpha(Z+2,A+4)\,Y(Z+2,A+4)}_{\alpha\text{ decay}} + \underbrace{\sum_j y_j(Z,A)\,R_f(j)}_{\text{Fission}} \\ &- \underbrace{\lambda_{\mathrm{tot}}(Z,A)\,Y(Z,A)}_{\text{Destruction due to decays}}\,. \end{aligned} \tag{3.18}$$

Here $\lambda_{n\gamma}(Z,A) = \rho N_A \langle\sigma v\rangle_{n\gamma}(Z,A)\,Y_n = n_n \langle\sigma v\rangle_{n\gamma}(Z,A)$ is the neutron-capture rate per target nucleus, $\lambda_{\gamma n}$ the photodisintegration rate obtained from it through Eq. 3.16, λ_β and λ_α the β^- and α decay rates, and P_{kn} the probability that the β^- decay is followed by the emission of k neutrons. The fission rate of nucleus j per baryon, summed over the spontaneous, neutron-induced and β-delayed modes, is $R_f(j) = [\lambda_{\mathrm{sf}}(j) + \lambda_{nf}(j) + P_{\beta f}(j)\,\lambda_\beta(j)]\,Y_j$, with $\lambda_{nf} = \rho N_A \langle\sigma v\rangle_{nf} Y_n$, and $y_j(Z,A)$ is the number of fragments (Z,A) produced per fission of j. The emission probabilities satisfy $\sum_k P_{kn} + P_{\beta f} = 1$, and the total destruction rate is $\lambda_{\mathrm{tot}} = \lambda_{n\gamma} + \lambda_{\gamma n} + \lambda_\beta + \lambda_\alpha + \lambda_{\mathrm{sf}} + \lambda_{nf}$. The free neutrons obey

$$\frac{\mathrm{d}Y_n}{\mathrm{d}t} = \underbrace{\sum_{Z,A}\left[\lambda_{\gamma n} + \lambda_\beta \sum_k k P_{kn}\right] Y(Z,A) + \sum_j \nu_j\,R_f(j)}_{\text{Creation/Emission}} - \underbrace{\left(\frac{Y_n}{\tau_n} + \sum_{Z,A}\left[\lambda_{n\gamma} + \lambda_{nf}\right] Y(Z,A)\right)}_{\text{Destruction/Absorption}}, \tag{3.19}$$

where ν_j is the average number of neutrons emitted per fission and $\tau_n \simeq 878\,\mathrm{s}$ is the lifetime of the free neutron.

The structure of Eq. 3.18 already contains the essential ingredients of the r-process. The capture rate, $\lambda_{n\gamma}$, is proportional to the neutron number density $n_n = \rho N_A Y_n$. For reference

let us take $\langle\sigma v\rangle \sim 10^{-17}\,\mathrm{cm^3\,s^{-1}}$, a neutron density $n_n \sim 10^{24}\,\mathrm{cm^{-3}}$ that gives $\lambda_{n\gamma} \sim 10^7\,\mathrm{s^{-1}}$. At the same time, some neutron-rich nuclei decay with $\lambda_\beta \sim 1$–$10^3\,\mathrm{s^{-1}}$. Captures therefore proceed many orders of magnitude faster than β decays, which is the defining property of the r-process. Along an isotopic chain the captures continue until neutron emission rate is comparable with neutron capture rate. From there, matter advances to higher Z only by β decay (Sec. 3.3). Once the neutrons are exhausted[4], the neutron-capture rates and neutron induced fission rates stop ($\lambda_{n\gamma}, \lambda_{nf} \to 0$) and when the temperature is too low for photodisintegration ($\lambda_{\gamma n} \to 0$), only the decay terms remain. Equation 3.18 then becomes linear, $\mathrm{d}\mathbf{Y}/\mathrm{d}t = \mathsf{M}\,\mathbf{Y}$, as in Sec. 3.4.4, with a matrix M built from decay constants, branching ratios and fission yields alone. Its solution is a superposition of decay chains and describes the composition and the radioactive heating at the times of the kilonova (Sec. 3.4).

3.2 Nuclear statistical equilibrium and freeze-out

At temperatures $T \gtrsim 5$–$6\,\mathrm{GK}$, strong and electromagnetic reactions are much faster than the expansion of the ejecta [69]. So a type of equilibrium is reached. At that state each reaction is then balanced by its reverse rate given by detailed balance (Sec. 3.1.4). The abundances no longer depend on the individual rates and the network equation reduces to a set of algebraic equations that describe nuclear statistical equilibrium (NSE). Weak interactions are in general slower and not in equilibrium, so NSE does not determine Y_e. The electron fraction, therefore, enters as a parameter, set by the weak processes discussed in Chapter 2. In this section we derive the NSE composition, discuss its dependence on density, temperature and entropy, and describe how NSE breaks down as the ejecta cool, which fixes the initial conditions of the r-process.

3.2.1 The Saha equation

When all strong and electromagnetic reactions are in equilibrium, any nucleus can be built from free nucleons by a chain of reactions, each of which is balanced by its reverse. In terms of the chemical potentials we have the following:

$$\mu(Z,A) = \mu(Z,A-1) + \mu_n = \mu(Z-1,A-2) + \mu_p + \mu_n = \cdots = Z\mu_p + N\mu_n. \tag{3.20}$$

Where μ_p and μ_n are the chemical potentials of free protons and neutrons and $\mu(Z,A)$ is the chemical potential of the nucleus (Z,A). What we can see from Eq. 3.20 is that the chemical potential of any species in this regime is just the sum of the chemical potential of its nucleons. Now by inverting Eq. 3.15, the chemical potential of a non-degenerate species is

$$\mu_i = m_i c^2 + k_B T \ln\left(\frac{n_i}{g_i G_i}\left(\frac{2\pi\hbar^2}{m_i k_B T}\right)^{3/2}\right), \tag{3.21}$$

Inserting Eq. 3.21 for the nucleus and for the nucleons ($g_p = g_n = 2$, $G_p = G_n = 1$) into

[4] Neutron freeze-out occurs at $\sim 1\,\mathrm{s}$

Eq. 3.20 and defining the thermal wavelength $\Lambda_i = (2\pi\hbar^2/m_i k_B T)^{1/2}$ gives

$$n(Z,A) = \frac{g(Z,A)\,G(Z,A)}{\Lambda^3_{(Z,A)}} \left(\frac{n_p \Lambda_p^3}{2}\right)^Z \left(\frac{n_n \Lambda_n^3}{2}\right)^N \exp\left(\frac{B(Z,A)}{k_B T}\right), \tag{3.22}$$

where

$$B(Z,A) = [Z m_p + N m_n - m(Z,A)]\,c^2 \tag{3.23}$$

is the nuclear binding energy. Setting $m_p \simeq m_n \simeq m_u$ and $m(Z,A) \simeq A m_u$ in the prefactors[5] and dividing by $n_b = \rho N_A$, Eq. 3.22 becomes the Saha equation for the abundances [69, 77],

$$Y(Z,A) = g(Z,A)\,G(Z,A)\,\frac{A^{3/2}}{2^A}\,\theta^{A-1} \exp\left(\frac{B(Z,A)}{k_B T}\right) Y_p^Z\, Y_n^N, \tag{3.24}$$

with the dimensionless quantity

$$\theta \equiv n_b \Lambda_u^3 = \rho N_A \left(\frac{2\pi\hbar^2}{m_u k_B T}\right)^{3/2}, \tag{3.25}$$

Now we shall notice the similarity to the Eq. 3.16, and that is because Eq. 3.24 is detailed balance applied to every link of the chain that builds the nucleus. The full derivation is presented in Appendix B. For this reason NSE and the reverse rates of the network are mutually consistent only if both use the same binding energies. Coulomb interactions in the plasma lower the free energy of each nucleus and thereby increase its effective binding energy; this correction is equivalent to the screening of the charged-particle captures that build the nucleus [69].

The beauty of Eq. 3.24 is that it expresses every abundance in terms of two unknowns, Y_n and Y_p. They are fixed by the two constraints of Eq. 3.2, $\sum_i A_i Y_i = 1$ and $\sum_i Z_i Y_i = Y_e$. The NSE composition is therefore a function of (ρ, T, Y_e) alone and keeps no memory of how the matter reached equilibrium. Matter that has been heated above $\sim 6\,\mathrm{GK}$ starts its nucleosynthesis from a composition fixed by Y_e. However, fluid elements that never reach such temperatures retain their previous composition and cannot be described by NSE.

3.2.2 Composition in NSE: binding energy versus entropy

The structure of Eq. 3.24 makes explicit the competition that determines the composition. Taking the logarithm,

$$\ln Y(Z,A) = \frac{B(Z,A)}{k_B T} - (A-1)\ln\frac{1}{\theta} + Z\ln Y_p + N\ln Y_n + \ln\left[g\,G\,\frac{A^{3/2}}{2^A}\right]. \tag{3.26}$$

Binding A nucleons into a nucleus gains the energy $B(Z,A)$, but in a gas with $\theta \ll 1$ each nucleon that loses its translational freedom costs a factor of $k_B \ln(1/\theta)$ of entropy. For example, if in Eq. 3.26 we replace $A \to A+1$ its abundance will be reduced by a factor of

[5] These replacements multiply $Y(Z,A)$ by a factor of the form $c_p^Z c_n^N$, with $c_{p,n} = (m_u/m_{p,n})^{3/2} \simeq 0.99$, which is absorbed by the solution for Y_p and Y_n; they change the NSE composition only at the per cent level. The exact masses are kept in $B(Z,A)$.

θ, $Y(Z, A+1) = \theta Y(Z, A)$ assuming all the other contributors remain the same. So what we can learn from this is that in order to produce abundance of heavy elements, the terms such as the binding energy should contribute enough to reduce the effect of the θ. One can understand this from the Sackur–Tetrode entropy per baryon of free neutrons,

$$s = k_B Y_n \left[\ln\left(\frac{2}{Y_n \theta}\right) + \frac{5}{2} \right] \tag{3.27}$$

so $\ln(1/\theta)$ is the leading term in the entropy of a free nucleon in units of k_B. From these results we can define the temperature at which nuclei and free neutrons have comparable abundances by balancing the first two terms of Eq. 3.26

$$k_B T_* \simeq \frac{B(Z,A)}{(A-1)\ln(1/\theta)} = \frac{B/A}{(1-1/A)\ln(1/\theta)}. \tag{3.28}$$

As the temperature rises at fixed density θ decreases (see Eq. 3.25), NSE therefore evolves from heavy nuclei to α particles and free nucleons, and finally to free nucleons alone, with $Y_n = 1 - Y_e$ and $Y_p = Y_e$. Lowering the density at fixed temperature, has the same effect. Below T_* the composition is dominated by the most bound nuclei compatible with the constraint $\sum_i Z_i Y_i = Y_e$. How neutron-rich these nuclei can become follows from the ratio of two neighbouring isotopes in Eq. 3.24,

$$\frac{Y(Z,A+1)}{Y(Z,A)} = \frac{g(Z,A+1)\,G(Z,A+1)}{2\,g(Z,A)\,G(Z,A)} \left(\frac{A+1}{A}\right)^{3/2} \theta\, Y_n \exp\left(\frac{S_n(Z,A+1)}{k_B T}\right), \tag{3.29}$$

with $S_n(Z, A+1) = B(Z, A+1) - B(Z, A)$ the neutron separation energy. Adding neutrons to a nucleus is favoured only while $S_n \gtrsim k_B T \ln[1/(\theta Y_n)]$, so the neutron excess that nuclei can hold is limited. At low Y_e the remaining neutrons stay free, and NSE at low temperature consists of neutron-rich nuclei coexisting with a large abundance of free neutrons. This is the origin of the neutron-to-seed ratio discussed in Sec. 3.3, where Eq. 3.29 reappears as the $(n,\gamma) \rightleftarrows (\gamma, n)$ equilibrium that defines the r-process path.

3.2.3 Quasi-equilibrium and freeze-out

As the ejecta expand and cool, the reaction rates fall at very different speeds, and NSE breaks down in stages. A reaction drops out of equilibrium when its timescale becomes longer than the expansion timescale, $\lambda(T(t))\,\tau_{\rm exp} \lesssim 1$.

The first link to break is the one between the light particles, n, p and α, and the heavy nuclei. There are no stable nuclei with $A = 5$ or $A = 8$ (^{8}Be decays into two α particles within $\sim 10^{-16}$ s), so nucleons and α are assembled into heavier nuclei through effective three-body reactions such as the triple-α reactions [78]. But their rates decrease rapidly as the density drops. Above this bottleneck, nuclei with $A \geq 12$ remain for some time in equilibrium with each other and with n, p and α through faster two-body reactions. This quasi-statistical equilibrium (QSE) [79] is described by Eq. 3.24 written relative to a reference nucleus (Z_r, A_r),

$$Y(Z,A) = \frac{C(Z,A)}{C(Z_r,A_r)}\,\theta^{A-A_r} \exp\left[\frac{B(Z,A) - B(Z_r,A_r)}{k_B T}\right] Y_p^{Z-Z_r}\, Y_n^{N-N_r}\, Y(Z_r,A_r), \tag{3.30}$$

with $C(Z,A) = g\,G\,A^{3/2}/2^A$. The difference with NSE is that $Y(Z_r, A_r)$, or equivalently the total abundance of heavy nuclei $Y_h = \sum_{A\geq 12} Y_i$, is no longer fixed by Eq. 3.24. The abundance of heavy nuclei in this regime will be then given by the amount of matter that was able to cross the bottleneck. If this flow is slow compared with the expansion, the heavy nuclei cannot absorb the α particles that NSE would bind into them, and the matter leaves equilibrium with a large α abundance. This effect is called an α-rich freeze-out.

The number of heavy nuclei produced in this phase can be estimated for neutron-rich matter [80]. Since ^{9}Be is disintegrated into $\alpha + \alpha + n$ by photons of only 1.573 MeV, at the relevant temperatures it is in equilibrium with these particles, and by Eq. 3.30 its abundance scales as $Y(^9\mathrm{Be}) \propto \theta^2 Y_\alpha^2 Y_n \propto \rho^2$. Each reaction $^9\mathrm{Be}(\alpha, n)^{12}\mathrm{C}$ creates one heavy nucleus, which subsequently grows by α captures into a seed, so that

$$\frac{\mathrm{d}Y_{\mathrm{seed}}}{\mathrm{d}t} \simeq \rho N_A \langle\sigma v\rangle_{\alpha n}(^9\mathrm{Be})\, Y_\alpha\, Y(^9\mathrm{Be}) \propto \rho^3. \tag{3.31}$$

For a radiation-dominated adiabat $\rho \propto T^3/s$, and the time spent in the temperature window of this α-process is proportional to τ_{exp}, so the number of seeds scales as

$$Y_{\mathrm{seed}} \propto \frac{\tau_{\mathrm{exp}}}{s^3}. \tag{3.32}$$

High entropy and fast expansion suppress the formation of seeds. This is the physical origin of the combination s^3/τ_{exp} that characterizes successful r-process conditions in neutrino-driven winds [80]. In the dynamical ejecta the entropy is low ($s \lesssim 10$–30), the bottleneck is not limiting, and the seeds are the neutron-rich heavy nuclei already present in NSE.

Finally, as the temperature drops to $T_9 \approx 2$–3, the Coulomb barriers stop the α captures on the seed nuclei, and all charged-particle reactions freeze out [78, 81]. From this point the proton number of a nucleus changes only through β decay, whereas neutron captures, which face no barrier, continue. The composition at charged-particle freeze-out, consisting of free neutrons, α particles and a distribution of seed nuclei with average charge $\bar{Z}$ and mass number $\bar{A}$, is the initial condition of the r-process described in the next section.

3.3 The r-process: neutron-to-seed ratio, reaction path and fission cycling

After charged-particle freeze-out the matter consists of free neutrons, seed nuclei and, depending on the entropy, α particles (Sec. 3.2.3). The r-process is the subsequent capture of the free neutrons by the seeds, interrupted by β decays that increase the nuclear charge, until the neutrons are exhausted. Its dynamics is contained in Eqs. 3.18 and 3.19; here we summarize the features needed to interpret the composition and the heating of the ejecta. Detailed reviews can be found in Refs. [1, 70, 82].

3.3.1 Neutron-to-seed ratio

The amount of heavy material that the r-process can build is measured by the neutron-to-seed ratio at its beginning, Y_n/Y_{seed} [81]. Each captured neutron increases the mass number of a

seed by one, so if all neutrons are captured and fission is negligible, the final average mass number of the heavy nuclei is

$$\langle A\rangle \simeq \frac{\sum_{seeds} A_i Y_i + Y_n}{\sum_{seeds} Y_i} \simeq \bar{A} + \frac{Y_n}{Y_{\rm seed}}, \tag{3.33}$$

where $\bar{A}$ and $\bar{Z}$ denote the average mass and charge numbers of the seeds. The constraints of Eq. 3.2 with only free neutrons and seeds read $Y_e = \bar{Z} Y_{\rm seed}$ and $1 = \bar{A} Y_{\rm seed} + Y_n$, so that

$$\frac{Y_n}{Y_{\rm seed}} = \frac{\bar{Z}}{Y_e} - \bar{A}, \qquad \langle A\rangle \simeq \frac{\bar{Z}}{Y_e}. \tag{3.34}$$

From the second equation we can estimate the number of baryons per seed using Y_e. For seeds with $\bar{Z} \approx 35$ one finds $\langle A\rangle \approx 117$ for $Y_e = 0.30$. Lanthanides ($57 \le Z \le 71$, $A \approx 140$–176) and the third peak are therefore produced only for $Y_e \lesssim 0.25$. For $Y_e \gtrsim 0.3$ the r-process does not proceed beyond the second peak [22]. This is the nuclear origin of the lanthanide-rich and lanthanide-poor kilonova components discussed in Chapter 1. At higher entropy the seeds are fewer, Eq. 3.32, and part of the protons are bound in α particles, so that the same neutron-to-seed ratio is reached at larger Y_e.

3.3.2 The reaction path and waiting points

At the beginning of the r-process the neutron density is large, and neutron captures are much faster than β decays (Sec. 3.1.5). If the temperature is high, $T_9 \gtrsim 1$, photodisintegrations are equally fast, and each isotopic chain reaches $(n,\gamma) \rightleftarrows (\gamma, n)$ equilibrium. The abundances along the chain are then given by Eq. 3.29 with $\theta Y_n = n_n \Lambda_u^3$, and the ratio $Y(Z, A{+}1)/Y(Z, A)$ falls below unity once the neutron separation energy drops below

$$S_n^0 = k_B T \ln \frac{2}{n_n \Lambda_u^3} \tag{3.35}$$

with the ratios of partition functions and of mass numbers set to one [82]. The abundance of each isotopic chain is therefore concentrated in one or two nuclei with $S_n \approx S_n^0$, which define the r-process path. Since the path follows from Eq. 3.29, its location in equilibrium depends on the nuclear masses but not on the neutron-capture rates.

In equilibrium, matter moves from one isotopic chain to the next only by β decay. Summing Eq. 3.18 over each chain and neglecting fission and α decays, the total abundance $Y(Z) = \sum_A Y(Z, A)$ obeys

$$\frac{{\rm d}Y(Z)}{{\rm d}t} = \bar{\lambda}_\beta(Z-1)\, Y(Z-1) - \bar{\lambda}_\beta(Z)\, Y(Z), \qquad \bar{\lambda}_\beta(Z) = \sum_A \lambda_\beta(Z, A)\, \frac{Y(Z, A)}{Y(Z)}. \tag{3.36}$$

If the flow reaches a steady state, $\bar{\lambda}_\beta(Z)\, Y(Z) \approx$ const, and the abundance accumulates in the chains with the slowest decays. This happens at the neutron shell closures $N = 50$, 82 and 126. Beyond a closed shell S_n drops abruptly, so the path stays at the magic neutron number for several values of Z, and these closed-shell nuclei, with their small Q_β, decay slowly. These waiting points produce the r-process abundance peaks, which after the decay

to stability appear at $A \approx 80$, 130 and 195 [70, 82]. The β-decay rates of the waiting-point nuclei set the shape of the abundance pattern and the duration of the r-process, which is of the order of a second.

Notice that the r-process described previously assumed that the temperature is high. But the temperatures within the dynamical ejecta can drop below $T_9 \approx 0.5$ due to the expansion while neutrons are still abundant. Photodisintegration, which depends on temperature, then becomes ineffective and the equilibrium is lost. The path is then determined by the competition between neutron capture and β decay, $\lambda_{n\gamma}(Z, A) \sim \lambda_\beta(Z, A)$, and depends on the individual neutron-capture rates and not only on the masses. This type of r-process is called cold r-process [83, 84]. The energy released by the reactions themselves can reheat the matter and restore the equilibrium in part of the ejecta.

3.3.3 Freeze-out and decay to stability

The r-process ends when the neutrons are exhausted and captures can no longer compete with β decays. During this freeze-out the composition is still modified as the nuclei decay towards stability. Since β^- decays conserve A, each isobaric chain decays into its stable end point, with lifetimes that grow from milliseconds near the path to days and years close to stability. Nuclei with $A > 209$ have no stable isobar and continue through α decays until they reach lead or bismuth. This slow decay of the freeze-out composition through many long-lived nuclei and decay chains powers the kilonova from hours to months (Sec. 3.4).

3.3.4 Fission

When the neutron-to-seed ratio carries matter beyond $A \approx 250$, which by Eq. 3.34 requires $Y_e \lesssim 0.15$, the r-process reaches nuclei with low fission barriers that split by neutron-induced, β-delayed or spontaneous fission. In the network, fission enters through the fission rates and the fragment distribution $y_j(Z, A)$ of Eq. 3.18. Those fissions return matter to intermediate mass numbers together with a few neutrons per event. If neutrons are still available, the fragments capture them and are carried back to the fissioning region. This process is called fission cycling and if nuclei do it enough times the final abundances between the second and third peaks become insensitive to the initial conditions [58]. This is the cause of the robustness of the r-process at low Y_e.

3.4 Radioactive energy release, decay chains and self-heating

The reactions of the network do not only change the composition: they convert nuclear binding energy into kinetic energy of the reaction products. During the r-process this energy heats the ejecta and modifies their temperature, which in turn changes the reaction rates. In this section we derive the nuclear energy generation rate from the thermodynamics of a fluid element, describe how the energy is shared among the decay products, and study how it couples back to the temperature (self-heating). We then discuss the late-time regime, in

which the heating is determined by decay chains. The deposition of the energy in the ejecta at the times of the kilonova (thermalization) is treated in Chapter 4.

3.4.1 Nuclear energy generation and the entropy equation

Consider a fluid element with energy per baryon $\mathcal{E}$, including the rest-mass energy, and specific volume $1/n_b$. Energy conservation reads [69]

$$\mathrm{d}\mathcal{E} + P\,\mathrm{d}\left(\frac{1}{n_b}\right) = \mathrm{d}q_{\rm ext}, \tag{3.37}$$

where P is the pressure and $\mathrm{d}q_{\rm ext}$ the energy per baryon exchanged with the surroundings. In the case of the ejecta $\mathrm{d}q_{\rm ext}$ will be the energy carried away by neutrinos. Nuclear reactions do not appear in Eq. 3.37, because they conserve the total energy, they only transform rest-mass energy into thermal energy. This transformation becomes explicit when $\mathcal{E}$ is written as a function of the entropy per baryon s (in units of k_B), the specific volume and the composition, through the Gibbs relation

$$\mathrm{d}\mathcal{E} = k_B T\,\mathrm{d}s - P\,\mathrm{d}\left(\frac{1}{n_b}\right) + \sum_i \mu_i\,\mathrm{d}Y_i + \mu_e\,\mathrm{d}Y_e, \tag{3.38}$$

where the chemical potentials include the rest mass and the last term accounts for the electrons. Photons do not contribute because $\mu_\gamma = 0$. Combining Eqs. 3.37 and 3.38 gives the evolution of the entropy [69, 72, 85],

$$k_B T\,\frac{\mathrm{d}s}{\mathrm{d}t} = \dot{q}_{\rm nuc} + \dot{q}_{\rm ext}, \qquad \dot{q}_{\rm nuc} \equiv -\sum_i \left(\mu_i + Z_i\mu_e\right)\frac{\mathrm{d}Y_i}{\mathrm{d}t}, \tag{3.39}$$

where $\dot{q}_{\rm nuc}$ is the nuclear heating rate per baryon. It is positive when the composition evolves towards species of lower chemical potential hence the minus sign.

The structure of $\dot{q}_{\rm nuc}$ can be better understood when it is written in terms of reactions. Let R_r be the net rate per baryon (forward minus reverse) of reaction r, so that $\mathrm{d}Y_i/\mathrm{d}t = \sum_r N_r^i R_r$ with the multiplicities of Eq. 3.12. Then

$$\dot{q}_{\rm nuc} = -\sum_i \left(\mu_i + Z_i\mu_e\right)\frac{\mathrm{d}Y_i}{\mathrm{d}t}, \tag{3.40}$$

$$\dot{q}_{\rm nuc} = -\sum_i \left(\mu_i + Z_i\mu_e\right)\sum_r N_r^i R_r, \tag{3.41}$$

$$\dot{q}_{\rm nuc} = \sum_r \left(-\sum_i \left(\mu_i + Z_i\mu_e\right) N_r^i\right) R_r, \tag{3.42}$$

$$\dot{q}_{\rm nuc} = \sum_r \mathcal{A}_r\,R_r, \qquad \mathcal{A}_r \equiv -\sum_i N_r^i\left(\mu_i + Z_i\mu_e\right), \tag{3.43}$$

now we can see that $\mathcal{A}_r$ is the free energy released per reaction. For a reaction whose reverse is related to it by detailed balance (Sec. 3.1.4), Eq. 3.15 implies that the ratio of the reverse to the forward rate is $e^{-\mathcal{A}_r/k_BT}$, so that

$$R_r = R_r^{\rightarrow}\left(1 - e^{-\mathcal{A}_r/k_BT}\right), \tag{3.44}$$

with $R_r^{\rightarrow}$ the forward rate per baryon. Each term $\mathcal{A}_r R_r$ is therefore non-negative, as required by the second law, and it vanishes for reactions in equilibrium, $\mathcal{A}_r = 0$. One interesting result here is that reactions in equilibrium produce no entropy even though they change the composition as the equilibrium shifts. Only reactions out of equilibrium heat the matter [69]. In the r-process these are essentially the β decays and fission.

Let us now write $\dot{q}_{\rm nuc}$ in terms of Q-values. To do that first we write the chemical potential of each nucleus as in Eq. 3.21, $\mu_i = m_i c^2 + k_B T \eta_i$ with $\eta_i = \ln[n_i \Lambda_i^3/(g_i G_i)]$, and express the nuclear mass through the tabulated atomic mass excess Δ_i, $m_i c^2 = A_i m_u c^2 + \Delta_i - Z_i m_e c^2$ [69]. Then $\dot{q}_{nuc}$ from Eq. 3.39 is given as

$$\dot{q}_{nuc} = -\sum_i \left(A_i m_u c^2 + \Delta_i - Z_i m_e c^2 + k_B T \eta_i + Z_i \mu_e\right) \frac{{\rm d}Y_i}{{\rm d}t} \tag{3.45}$$

Since $\sum_i A_i \, {\rm d}Y_i/{\rm d}t = 0$, the rest mass of the baryons drops out and $\sum_i Z_i {\rm d}Y_i/{\rm d}t = {\rm d}Y_e/{\rm d}t$, therefore

$$\dot{q}_{\rm nuc} = -\sum_i \Delta_i \frac{{\rm d}Y_i}{{\rm d}t} - k_B T \sum_i \eta_i \frac{{\rm d}Y_i}{{\rm d}t} - \left(\mu_e - m_e c^2\right) \frac{{\rm d}Y_e}{{\rm d}t}. \tag{3.46}$$

The first term is the energy generation from the change in binding energy, the Q-values of all reactions weighted with their net rates. The second and third terms are thermodynamic corrections, the change of the translational and internal entropy of the nuclei and the change of the energy of the electron gas when Y_e changes. They are essential near equilibrium, where they cancel the first term, but negligible when k_BT and $\mu_e - m_e c^2$ are small compared with the Q-values. In that limit only decays remain, $\mathcal{A}_r \to Q_r$, and the energy generation rate per unit mass, $\dot{\epsilon} = N_A \dot{q}_{\rm nuc}$, becomes

$$\dot{\epsilon}(t) = N_A \sum_i \sum_x Q_{i,x} \, \lambda_{i,x} \, Y_i(t), \tag{3.47}$$

where x labels the decay modes of nucleus i. With $Y_i = X_i/A_i$ and $m_i \simeq A_i m_u$, Eq. 3.47 is Eq. 2.20 of Chapter 2.

3.4.2 Energy partition and neutrino losses

The energy $Q_{i,x}$ of a decay is shared among its products, and only part of it is deposited in the ejecta. In a β^- decay it is divided between the electron, the antineutrino and the γ rays (or neutrons) emitted in the de-excitation of the daughter nucleus. In an α decay it goes almost entirely into the kinetic energy of the α particle, and in fission mostly into the kinetic energy of the two fragments, $\approx 200\,{\rm MeV}$ per event. The ejecta are transparent to neutrinos at densities $\rho \ll 10^{11}\,{\rm g\,cm^{-3}}$, so the neutrino energy is lost. The corresponding term in Eq. 3.39 is

$$\dot{q}_{\rm ext} = -\sum_i \langle E_\nu \rangle_i \, \lambda_{\beta,i} \, Y_i = -\chi_\nu \sum_i Q_{\beta,i} \, \lambda_{\beta,i} \, Y_i, \tag{3.48}$$

where $\langle E_\nu \rangle_i$ is the average antineutrino energy per decay, known from experiment for measured decays and from theory otherwise, and χ_ν is the average fraction of the decay energy carried by the neutrino. For the neutron-rich nuclei of the r-process $\chi_\nu \approx 0.3$ is typical. This is less than the roughly equal sharing of the lepton energy between electron and antineutrino in a decay to the ground state, because the decays mainly populate excited states of the daughter so part of the Q_β is released in form of γ or neutron emission. Energy can also be gained by the absorption of neutrinos emitted by the merger remnant only if the ejecta is close enough to the remnant.

At early times the ejecta are dense enough to absorb the electrons, γ rays, α particles and fission fragments locally, and all the energy not carried by neutrinos heats the matter. At the times of the kilonova this is no longer true. γ rays escape after about a day, and the charged particles thermalize with an efficiency that decreases with the density (Chapter 2). The energy release must therefore be kept resolved by decay channel and decay product, $\dot{\epsilon}_x(t)$, which allows us to define the effective heating rate $\dot{\varepsilon}(t)$ as

$$\dot{\varepsilon}(t) = \sum_x f_x(t)\dot{\epsilon}_x(t) = f_\alpha \dot{\epsilon}_\alpha(t) + f_\beta(t)\dot{\epsilon}_\beta(t) + f_{fiss}\dot{\epsilon}_{fiss}(t), \tag{3.49}$$

where $f_x(t)$ is given as

$$f_\alpha(t) \simeq \left(1 + \frac{t}{t_\alpha}\right)^{-n}, \tag{3.50}$$

$$f_\beta(t) \simeq \chi_e \left(1 + \frac{t}{t_e}\right)^{-n} + \chi_\gamma \left(1 - \exp\left[-\frac{t_\gamma^2}{t^2}\right]\right), \tag{3.51}$$

$$f_{fiss}(t) \simeq \left(1 + \frac{t}{t_{fiss}}\right)^{-n}, \tag{3.52}$$

Where t_α, t_e, t_γ, and t_{fiss} are the time scale of the thermalization based on the products of the decay. χ_e and χ_γ are the average energy fraction for electrons and γ.

3.4.3 Self-heating

Equation 3.39 couples the nuclear reactions to the thermodynamic state of the fluid element. In a network calculation with self-heating, the density history $\rho(t)$ is taken from the hydrodynamics, the entropy evolves according to Eq. 3.39, and the temperature is recovered from the equation of state, $s = s(\rho, T, \{Y_i\})$. This approach, therefore, neglects the effect of the released energy on the dynamics. That means that it is assumed that the outer layers are not considerably accelerated through the radiation pressure of the inner layers. The approximation is justified by the energy budget. During the first 1.3 s Eq. 2.25 predicts a constant $\dot{\epsilon}_0 = 2 \times 10^{18}\,\mathrm{erg\,g^{-1}\,s^{-1}}$, which gives a total energy per gram of $\approx 2.6 \times 10^{18}\,\mathrm{erg\,g^{-1}}$, $\approx 2.7\,\mathrm{MeV}$ per baryon. This is small compared with the kinetic energy of the ejecta, $\approx 19\,\mathrm{MeV}$ per baryon at $v = 0.2c$, although it can modify the expansion of the slowest ejecta [86].

Temperature equation. Differentiating $s(\rho, T, \{Y_i\})$ and using Eq. 3.39,

$$c_V \frac{\mathrm{d}\ln T}{\mathrm{d}t} = \frac{\dot{q}_{\mathrm{nuc}} + \dot{q}_{\mathrm{ext}}}{k_B T} - \left(\frac{\partial s}{\partial \ln \rho}\right)_{T,Y} \frac{\mathrm{d}\ln \rho}{\mathrm{d}t} - \sum_i \left(\frac{\partial s}{\partial Y_i}\right)_{\rho,T} \frac{\mathrm{d}Y_i}{\mathrm{d}t}, \tag{3.53}$$

where $c_V \equiv (\partial s/\partial \ln T)_{\rho,Y}$ is the heat capacity per baryon in units of k_B. The three terms describe the nuclear heating, the adiabatic cooling by the expansion ($\partial s/\partial \ln \rho < 0$ and $\mathrm{d}\ln\rho/\mathrm{d}t < 0$), and the redistribution of the entropy among the species when the composition changes. However, to calculate the temperature from Eq. 3.53 we need the other terms in the equation, which are set by the EOS. Now the difficult task is to find the EOS.

For example, if we assume a radiation dominated regime, $s \propto T^3/\rho$ gives $c_V = 3s$, $(\partial s/\partial \ln \rho)_T = -s$ and no dependence on the composition, so that

$$\frac{\mathrm{d}\ln T}{\mathrm{d}t} = \frac{1}{3}\frac{\mathrm{d}\ln \rho}{\mathrm{d}t} + \frac{\dot{q}}{3\, s\, k_B T}, \tag{3.54}$$

with $\dot{q} = \dot{q}_{\mathrm{nuc}} + \dot{q}_{\mathrm{ext}}$. Without heating this is the adiabatic law $T \propto \rho^{1/3} \propto t^{-1}$. However, in many cases this assumption is not fulfilled. Fortunately for us, codes such as WinNet and SkyNet implement a modified Timmes EOS presented in Refs. [87, 88], which have three parts, a photon gas, an arbitrarily degenerate and relativistic electron/positron gas, and a non-degenerate, non-relativistic Boltzmann gas for the heavy ions. Such EOS are valid in a wider range so consistency is preserved.

3.4.4 Decay chains

Now that we know how to evolve the thermodynamic part consistently with the network equation, we should move to the details of the decays itself. We should remember that after the r-process freezes out, the network reduces to the linear system $\mathrm{d}\mathbf{Y}/\mathrm{d}t = \mathsf{M}\,\mathbf{Y}$ (Sec. 3.1.5). If the nuclei are ordered by decreasing atomic mass, every decay leads from a nucleus to lighter ones further down the list, because $Q > 0$ for all decay modes ($\beta^{\pm}$, electron capture, β-delayed neutron emission, α decay, spontaneous fission). The first species will be the heaviest and since there is no more synthesis it can only decay, so $\dot{Y}_1 = -\lambda_1 Y_1$. As consequence the first row of M is $(-\lambda_1, 0, \ldots, 0)$. If we do this for any nuclei at position i then the ith row will be something like $(c_1, c_2, \ldots, -\lambda_i, 0, \ldots, 0)$. M is therefore lower triangular. Its eigenvalues are its diagonal elements, $-\lambda_i$, with λ_i the total decay constants, and every abundance is a superposition of exponentials, $Y_i(t) = \sum_j c_{ij}\, e^{-\lambda_j t}$. For a linear chain $1 \to 2 \to \cdots \to n$ in which only the head is initially present, the coefficients are given by the Bateman solution,

$$Y_n(t) = Y_1(0) \left(\prod_{k=1}^{n-1} \lambda_k\right) \sum_{k=1}^{n} \frac{e^{-\lambda_k t}}{\prod_{j\neq k}(\lambda_j - \lambda_k)}, \tag{3.55}$$

The physics of a chain is already contained in its first two members,

$$Y_2(t) = Y_1(0)\, \frac{\lambda_1}{\lambda_2 - \lambda_1} \left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right) + Y_2(0)\, e^{-\lambda_2 t}. \tag{3.56}$$

If the daughter is shorter-lived than the parent, $\lambda_2 > \lambda_1$, then for $t \gg 1/(\lambda_2 - \lambda_1)$ it reaches transient equilibrium,

$$\lambda_2 Y_2 = \frac{\lambda_2}{\lambda_2 - \lambda_1}\, \lambda_1 Y_1 \xrightarrow{\lambda_2 \gg \lambda_1} \lambda_1 Y_1, \tag{3.57}$$

the last limit being secular equilibrium. The daughter decays at the rate at which it is produced, which is established by the half-life of the parent. A chain of short-lived nuclei headed by a long-lived one therefore releases its total energy at the decay rate of the head,

$$\dot{\epsilon}_{\rm chain}(t) \simeq N_A\,\lambda_1 Y_1(t) \sum_k Q_k^{\rm dep}, \tag{3.58}$$

where the sum runs over all members in equilibrium and $Q_k^{\rm dep}$ excludes the neutrino energy. If instead the daughter lives longer, $\lambda_2 < \lambda_1$, it accumulates and decays later with its own half-life. In both cases the late-time heating of a chain is paced by its longest-lived member, while its magnitude is set by the energy of all the members.

This mechanism is responsible for the most prominent features of the late-time heating [31]. Nuclei with $A > 209$ reach stability through α decays, which change A by four, and β^- decays, which do not change it. The quantity A mod 4 is therefore conserved, and the heavy nuclei decay along four independent series, $A = 4n$, $4n+1$, $4n+2$ and $4n+3$. From days to weeks after the merger each series is dominated by one relatively long-lived member followed by a fast sequence of α and β^- decays: ^{224}Ra ($T_{1/2} = 3.6\,$d), ^{225}Ac (10.0 d), ^{222}Rn (3.8 d) and ^{223}Ra (11.4 d), respectively [31, 32]. Each of them releases ≈ 20–$30\,$MeV through its chain, mostly as kinetic energy of α particles, which is much more than in a typical β decay and in a form that thermalizes efficiently. Their abundances are fixed at freeze-out by the flow of matter into the corresponding mass chains, including the α and β^- decays of heavier, mostly unmeasured nuclei, whose rates determine the branchings and the timing of the feeding. One special case is the spontaneous fission of ^{254}Cf ($T_{1/2} = 60.5\,$d), which adds $\approx 200\,$MeV per event on similar timescales.

Chapter 4

Kilonova Light Curves

4.1 Li-Paczyński model

4.2 Radiative transfer and the diffusion regime

4.3 Opacities of r-process matter

4.4 Light-curve modeling

Chapter 5

Methodology on the impact of theoretical decay rates across thermodynamics trajectories

5.1 Binary neutron-star simulation

5.2 From simulation to trajectories: ejecta extraction and expansion

5.3 Initial termodinamic trajectories treatment

5.4 Nuclear reaction network setup and theoretical reaction reates

5.5 Modified Li-Paczyński model

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Chapter 6

Results

6.1 Nucleosynthesis and heating rates

6.1.1 Relevant α and β

6.2 Kilonova light curves

6.3 Detectability with current and future telescopes

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Chapter 7

Discussion

7.1 Comparison with previous studies

7.2 Limitations and model assumptions

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Chapter 8

Conclusions and Outlook

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Appendix A

Quadrupole and L_{GW}

In this appendix we will go through the details of finding the mass quadrupole components, explain the approximations assumed and finding the L_{GW}. Lets start with the definitions, as we saw from Fig. 2.2 the position are:

$$\mathbf{r}_1 = -\frac{\mu a}{m_1}(cos(\theta), sin(\theta), 0), \tag{A.1}$$

$$\mathbf{r}_2 = \frac{\mu a}{m_2}(cos(\theta), sin(\theta), 0). \tag{A.2}$$

$$\tag{A.3}$$

We used $||\mathbf{r}_1|| = m_2 a/(m_1 + m_2) = \mu a/m_1$ with $\mu = m_1 m_2/(m_1 + m_2)$. The first quadrupole component will be:

$$Q_{xx} = m_1 r_{x1} r_{x1} + m_2 r_{x2} r_{x2} = m_1 \frac{\mu^2 a^2}{m_1^2} cos^2(\theta) + m_2 \frac{\mu^2 a^2}{m_2^2} cos^2(\theta), \tag{A.4}$$

$$Q_{xx} = \mu^2 a^2 cos^2(\theta)(\frac{1}{m_1} + \frac{1}{m_2}), \tag{A.5}$$

$$Q_{xx} = \mu a^2 cos^2(\theta), \tag{A.6}$$

$$Q_{xx} = \frac{\mu a^2}{2}(cos(2\theta) + 1). \tag{A.7}$$

$$\tag{A.8}$$

Similarly for Q_{yy}:

$$Q_{yy} = m_1 r_{y1} r_{y1} + m_2 r_{y2} r_{y2} = m_1 \frac{\mu^2 a^2}{m_1^2} sin^2(\theta) + m_2 \frac{\mu^2 a^2}{m_2^2} sin^2(\theta), \tag{A.9}$$

$$Q_{yy} = \mu^2 a^2 sin^2(\theta)(\frac{1}{m_1} + \frac{1}{m_2}), \tag{A.10}$$

$$Q_{yy} = \mu a^2 sin^2(\theta), \tag{A.11}$$

$$Q_{yy} = \frac{\mu a^2}{2}(-cos(2\theta) + 1). \tag{A.12}$$

$$\tag{A.13}$$

Finally for Q_{xy}

$$Q_{xy} = m_1 r_{x1} r_{y1} + m_2 r_{x2} r_{y2} = m_1 \frac{\mu^2 a^2}{m_1^2} sin(\theta) cos(\theta) + m_2 \frac{\mu^2 a^2}{m_2^2} sin(\theta) cos(\theta), \tag{A.14}$$

$$Q_{xy} = \mu^2 a^2 sin(\theta) cos(\theta) (\frac{1}{m_1} + \frac{1}{m_2}), \tag{A.15}$$

$$Q_{xy} = \mu a^2 sin(\theta) cos(\theta), \tag{A.16}$$

$$Q_{xy} = \frac{\mu a^2}{2} sin(2\theta). \tag{A.17}$$

$$\tag{A.18}$$

Now we need the third derivate with respect to time. We will assume two things, $\ddot{\theta} << 1$ and $\dot{a} << 1$. Effectively we are saying that the change in angular velocity and the change in the radius is slow which is referred as an adiabatic process. To see how good the approximation is we can compare the orbit time scale with the inspiral timescale. Which in fact is the situation, therefore is a good approximation until the merger itself. Therefore it is easy to see that:

$$\dddot{Q}_{xx} = 4\mu a^2 \dot{\theta}^3 sin(2\theta) \tag{A.19}$$

$$\dddot{Q}_{yy} = -4\mu a^2 \dot{\theta}^3 sin(2\theta) \tag{A.20}$$

$$\dddot{Q}_{xy} = -4\mu a^2 \dot{\theta}^3 cos(2\theta) \tag{A.21}$$

$$\tag{A.22}$$

Now we meed to calculate the following term remembering that we are using einstein notation for the repeated indices and that $Q_{xy} = Q_{yx}$:

$$\dddot{Q}_{ij} \dddot{Q}_{ij} - \dddot{Q}_{ii} \dddot{Q}_{jj} = \dddot{Q}_{xx} \dddot{Q}_{xx} + \dddot{Q}_{xy} \dddot{Q}_{xy} + \dddot{Q}_{yx} \dddot{Q}_{yx} + \dddot{Q}_{yy} \dddot{Q}_{yy} \tag{A.23}$$

$$- \dddot{Q}_{xx} \dddot{Q}_{xx} - \dddot{Q}_{yy} \dddot{Q}_{xx} - \dddot{Q}_{xx} \dddot{Q}_{yy} - \dddot{Q}_{yy} \dddot{Q}_{yy} \tag{A.24}$$

$$= \dddot{Q}_{xx}^2 + 2\dddot{Q}_{xy}^2 + \dddot{Q}_{yy}^2 - \dddot{Q}_{xx}^2 - 2\dddot{Q}_{yy} \dddot{Q}_{xx} - \dddot{Q}_{yy}^2 \tag{A.25}$$

$$= \dddot{Q}_{xx}^2 + 2\dddot{Q}_{xy}^2 + \dddot{Q}_{yy}^2 - (\dddot{Q}_{xx} + \dddot{Q}_{yy})^2 \tag{A.26}$$

$$= 16\mu^2 a^4 \dot{\theta}^6 (sin^2(2\theta) + 2cos^2(2\theta) + sin^2(2\theta)) \tag{A.27}$$

$$= 32\mu^2 a^4 \dot{\theta}^6 = \frac{32 G^3 m_1^2 m_2^2 M}{a^5} \tag{A.28}$$

Finally:

$$L_{GW} = \frac{G}{5c^5} (\dddot{Q}_{ij} \dddot{Q}_{ij} - \dddot{Q}_{ii} \dddot{Q}_{jj}) = \frac{32 G^4 m_1^2 m_2^2 M}{c^5 a^5} \tag{A.29}$$

Appendix B

Derivation of the Saha equation of nuclear statistical equilibrium

This appendix derives the Saha equation, Eq. 3.24, used in Sec. 3.2.1. It is obtained by assuming a Maxwell-Boltzmann distribution and then using the equilibrium condition $\mu_{Z,A} = Z\mu_p + N\mu_n$ (Secs. B.1–B.3).

B.1 Ideal non-degenerate gas of particles with internal structure

Consider species i of mass m_i in a volume V at temperature T. A particle of this species is labelled by its momentum $\mathbf{p}$ and by its internal state l. The state l has excitation energy E_l and spin J_l, and $l = 0$ is the ground state. In the non-relativistic limit the single-particle energy, including the rest mass, is

$$\varepsilon_{\mathbf{p},l} = m_i c^2 + E_l + \frac{p^2}{2m_i}, \tag{B.1}$$

and each level has the magnetic degeneracy $2J_l + 1$. In the grand-canonical ensemble with chemical potential μ_i, which here includes the rest mass, the mean occupation of a single-particle state is

$$\langle n_{\mathbf{p},l} \rangle = \frac{1}{\exp\left[(\varepsilon_{\mathbf{p},l} - \mu_i)/k_B T\right] \pm 1}, \tag{B.2}$$

with the upper sign for fermions and the lower for bosons. If $\exp[(\varepsilon_{\mathbf{p},l} - \mu_i)/k_B T] \gg 1$, both statistics reduce to the Maxwell–Boltzmann,

$$\langle n_{\mathbf{p},l} \rangle \simeq \exp[(\mu_i - \varepsilon_{\mathbf{p},l})/k_B T]. \tag{B.3}$$

The number density is the sum over states times the occupation of the state.

$$n_i = \int \frac{d^3\mathbf{P}}{(2\pi\hbar)^3} \sum_l (2J_l+1)\langle n_{\mathbf{p},l}\rangle, \tag{B.4}$$

$$n_i = \int_0^\infty \frac{4\pi p^2 dp}{(2\pi\hbar)^3} \sum_l (2J_l+1) \exp\left[\frac{1}{k_B T}(\mu_i - m_i c^2 - E_l - \frac{p^2}{2m_i})\right], \tag{B.5}$$

$$n_i = \exp\left[\frac{1}{k_B T}(\mu_i - m_i c^2)\right]\left[\sum_l (2J_l+1)\,\exp\left(-\frac{E_l}{k_B T}\right)\right]\int_0^\infty dp \frac{p^2}{2\pi^2\hbar^3}\,\exp\left[-\frac{p^2}{2m_i k_B T}\right]. \tag{B.6}$$

The momentum integral is a Gaussian,

$$\int_0^\infty dp \frac{p^2}{2\pi^2\hbar^3}\,\exp\left[-\frac{p^2}{2m_i k_B T}\right] = \left[\frac{m_i k_B T}{2\pi\hbar^2}\right]^{\frac{3}{2}}. \tag{B.7}$$

From that we define the thermal wavelength of Sec. 3.2.1.

$$\left(\frac{m_i k_B T}{2\pi\hbar^2}\right)^{3/2} = \left(\frac{1}{\Lambda_i}\right)^3, \tag{B.8}$$

The internal sum is $g_i G_i(T)$, with $g_i = 2J_0 + 1$ and G_i the partition function of Eq. 3.6. Therefore

$$n_i = \frac{g_i G_i(T)}{\Lambda_i^3}\,\exp\left(\frac{\mu_i - m_i c^2}{k_B T}\right), \tag{B.9}$$

which is Eq. 3.15. Its inverse is the chemical potential of Eq. 3.21,

$$\mu_i = m_i c^2 + k_B T \ln\left(\frac{n_i \Lambda_i^3}{g_i G_i(T)}\right). \tag{B.10}$$

B.2 The Saha equation for number densities

Inserting Eq. B.10 into both sides of $\mu_{Z,A} = Z\mu_p + N\mu_n$, with $g_p = g_n = 2$ and $G_p = G_n = 1$ for the nucleons (spin-$\frac{1}{2}$ particles without bound excited states), gives

$$m(Z,A)c^2 + k_B T \ln\frac{n(Z,A)\Lambda^3_{(Z,A)}}{g(Z,A)G(Z,A)} = Z m_p c^2 + N m_n c^2 + k_B T\left[Z\ln\frac{n_p\Lambda_p^3}{2} + N\ln\frac{n_n\Lambda_n^3}{2}\right]. \tag{B.11}$$

The rest masses combine into the binding energy of Eq. 3.23, $B(Z,A) = [Z m_p + N m_n - m(Z,A)]c^2$. Dividing by $k_B T$ and exponentiating,

$$n(Z,A) = \frac{g(Z,A)\,G(Z,A)}{\Lambda^3_{(Z,A)}}\left(\frac{n_p\Lambda_p^3}{2}\right)^Z\left(\frac{n_n\Lambda_n^3}{2}\right)^N \exp\left(\frac{B(Z,A)}{k_B T}\right), \tag{B.12}$$

which is Eq. 3.22. The masses in $B(Z,A)$ are nuclear masses, because in NSE the atoms are completely ionized and the electrons are counted separately.

B.3 From number densities to abundances

All thermal wavelengths are now expressed in terms of Λ_u, the thermal wavelength at the atomic mass unit. Exactly,

$$\Lambda_p^3 = c_p\,\Lambda_u^3, \qquad \Lambda_n^3 = c_n\,\Lambda_u^3, \qquad \Lambda_{(Z,A)}^{-3} = \epsilon_A\,A^{3/2}\,\Lambda_u^{-3}, \tag{B.13}$$

with

$$c_p = \left(\frac{m_u}{m_p}\right)^{3/2} = 0.9892, \qquad c_n = \left(\frac{m_u}{m_n}\right)^{3/2} = 0.9871, \qquad \epsilon_A = \left(\frac{m(Z,A)}{Am_u}\right)^{3/2}. \tag{B.14}$$

Substituting Eq. B.13 into Eq. B.12 and writing $n_i = n_b Y_i$,

$$n_b Y(Z,A) = \epsilon_A\,g\,G\,\frac{A^{3/2}}{\Lambda_u^3}\,\frac{(n_b\Lambda_u^3)^A}{2^A}\,(c_pY_p)^Z\,(c_nY_n)^N\exp\left(\frac{B}{k_BT}\right). \tag{B.15}$$

Dividing by n_b and collecting $(n_b\Lambda_u^3)^{A-1} = \theta^{A-1}$, with θ defined in Eq. 3.25, yields

$$Y(Z,A) = \epsilon_A\;g(Z,A)\,G(Z,A)\,\frac{A^{3/2}}{2^A}\,\theta^{A-1}\exp\left(\frac{B(Z,A)}{k_BT}\right)(c_pY_p)^Z\,(c_nY_n)^N. \tag{B.16}$$

With $\epsilon_A = c_p = c_n = 1$ this is Eq. 3.24.

References

[1] J. J. Cowan et al., Reviews of Modern Physics **93**, 015002 (2021).

[2] S. Rosswog and O. Korobkin, Annalen der Physik **536**, 2200306 (2024).

[3] B. D. Metzger et al., Monthly Notices of the Royal Astronomical Society **406**, 2650 (2010).

[4] J. Barnes et al., The Astrophysical Journal **829**, 110 (2016).

[5] Y.-H. Yang et al., Nature **626**, 742 (2024).

[6] R. H. Cyburt et al., Reviews of Modern Physics **88**, 015004 (2016).

[7] F.-K. Thielemann et al., Space Science Reviews **214**, 62 (2018).

[8] E. M. Burbidge et al., Reviews of Modern Physics **29**, 547 (1957).

[9] F. Käppeler et al., Reviews of Modern Physics **83**, 157 (2011).

[10] F.-K. Thielemann and J. J. Cowan, The European Physical Journal A **62**, 105 (2026).

[11] R. Hoffman, S. Woosley, and Y.-Z. Qian, Nuclear Physics A **621**, 397 (1997).

[12] S. E. Woosley et al., The Astrophysical Journal **433**, 229 (1994).

[13] A. Arcones, H.-T. Janka, and L. Scheck, Astronomy & Astrophysics **467**, 1227 (2007).

[14] C. Sneden, J. J. Cowan, and R. Gallino, Annual Review of Astronomy and Astrophysics **46**, 241 (2008).

[15] A. P. Ji et al., The Astrophysical Journal **830**, 93 (2016).

[16] A. Wallner et al., Nature Communications **6**, 5956 (2015).

[17] J. M. Lattimer and D. N. Schramm, The Astrophysical Journal **210**, 549 (1976).

[18] L.-X. Li and B. Paczyński, The Astrophysical Journal **507**, L59 (1998).

[19] D. Kasen, N. R. Badnell, and J. Barnes, The Astrophysical Journal **774**, 25 (2013).

[20] M. Tanaka and K. Hotokezaka, The Astrophysical Journal **775**, 113 (2013).

[21] J. Barnes and D. Kasen, The Astrophysical Journal **775**, 18 (2013).

[22] J. Lippuner and L. F. Roberts, The Astrophysical Journal **815**, 82 (2015).

[23] B. P. Abbott et al., Physical Review Letters **119**, 161101 (2017).

[24] D. A. Coulter et al., Science **358**, 1556 (2017).

[25] M. R. Drout et al., Science **358**, 1570 (2017).

[26] V. A. Villar et al., The Astrophysical Journal Letters **851**, L21 (2017).

[27] B. D. Metzger, Living Reviews in Relativity **20**, 3 (2017).

[28] E. Waxman et al., Monthly Notices of the Royal Astronomical Society **481**, 3423 (2018).

[29] R. T. Wollaeger et al., Monthly Notices of the Royal Astronomical Society **478**, 3298 (2018).

[30] G. Ricigliano et al., Monthly Notices of the Royal Astronomical Society **529**, 647 (2024).

[31] M.-R. Wu et al., Physical Review Letters **122**, 062701 (2019).

[32] M.-H. Chen and E.-W. Liang, Monthly Notices of the Royal Astronomical Society **527**, 5540 (2023).

[33] A. J. Levan et al., Nature **626**, 737 (2024).

[34] J. H. Gillanders and S. J. Smartt, Monthly Notices of the Royal Astronomical Society **538**, 1663 (2025).

[35] Z. Liu, J.-a. Jiang, and W. Zhao, The Astrophysical Journal Letters **983**, L34 (2025).

[36] D. Hernandez Porras et al., Journal of High Energy Astrophysics **56**, 100761 (2027).

[37] C. J. Horowitz, J. Piekarewicz, and B. Reed, Physical Review C **102**, 044321 (2020).

[38] G. Baym, H. A. Bethe, and C. J. Pethick, Nuclear Physics A **175**, 225 (1971).

[39] F. Özel and P. Freire, Annual Review of Astronomy and Astrophysics **54**, 401 (2016).

[40] T. E. Riley et al., The Astrophysical Journal Letters **918**, L27 (2021).

[41] J. R. Oppenheimer and G. M. Volkoff, Physical Review **55**, 374 (1939).

[42] R. C. Tolman, Physical Review **55**, 364 (1939).

[43] K. Hebeler et al., The Astrophysical Journal **773**, 11 (2013).

[44] A. Kurkela et al., The Astrophysical Journal **789**, 127 (2014).

[45] G. Baym et al., Reports on Progress in Physics **81**, 056902 (2018).

[46] E. Fonseca et al., The Astrophysical Journal Letters **915**, L12 (2021).

[47] G. Raaijmakers et al., The Astrophysical Journal Letters **918**, L29 (2021).

[48] M. Alford et al., The Astrophysical Journal **629**, 969 (2005).

[49] B. Paczynski, Symposium - International Astronomical Union **73**, 75 (1976).

[50] T. M. Tauris, N. Langer, and P. Podsiadlowski, Monthly Notices of the Royal Astronomical Society **451**, 2123 (2015).

[51] T. M. Tauris et al., The Astrophysical Journal **846**, 170 (2017).

[52] P. C. Peters and J. Mathews, Physical Review **131**, 435 (1963).

[53] D. Radice et al., Monthly Notices of the Royal Astronomical Society **460**, 3255 (2016).

[54] D. Radice et al., The Astrophysical Journal **869**, 130 (2018).

[55] D. Radice, S. Bernuzzi, and A. Perego, Annual Review of Nuclear and Particle Science **70**, 95 (2020).

[56] C. J. Krüger and F. Foucart, Physical Review D **101**, 103002 (2020).

[57] M. Shibata and K. Hotokezaka, Annual Review of Nuclear and Particle Science **69**, 41 (2019).

[58] O. Korobkin et al., Monthly Notices of the Royal Astronomical Society **426**, 1940 (2012).

[59] B. P. Abbott et al., The Astrophysical Journal Letters **848**, L12 (2017).

[60] B. P. Abbott et al., Physical Review Letters **119**, 161101 (2017).

[61] R. D. Blandford and R. L. Znajek, Monthly Notices of the Royal Astronomical Society **179**, 433 (1977).

[62] S. E. Woosley, The Astrophysical Journal **405**, 273 (1993).

[63] A. I. MacFadyen and S. E. Woosley, The Astrophysical Journal **524**, 262 (1999).

[64] T. Ahumada et al., Nature Astronomy **5**, 917 (2021).

[65] K. Hotokezaka, R. Sari, and T. Piran, Monthly Notices of the Royal Astronomical Society **468**, 91 (2017).

[66] K. Hotokezaka et al., Monthly Notices of the Royal Astronomical Society **459**, 35 (2016).

[67] D. Kasen and J. Barnes, The Astrophysical Journal **876**, 128 (2019).

[68] M. Tanaka et al., Monthly Notices of the Royal Astronomical Society **496**, 1369 (2020).

[69] M. Reichert et al., The Astrophysical Journal Supplement Series **268**, 66 (2023).

[70] M. Arnould, S. Goriely, and K. Takahashi, Physics Reports **450**, 97 (2007).

[71] W. R. Hix and F.-K. Thielemann, Journal of Computational and Applied Mathematics **109**, 321 (1999).

[72] J. Lippuner and L. F. Roberts, The Astrophysical Journal Supplement Series **233**, 18 (2017).

[73] D. F. Rojas-Gamboa, N. G. Kelkar, and O. L. Caballero, Physical Review C **110**, 035804 (2024).

[74] W. A. Fowler, G. R. Caughlan, and B. A. Zimmerman, Annual Review of Astronomy and Astrophysics **5**, 525 (1967).

[75] T. Rauscher and F.-K. Thielemann, Atomic Data and Nuclear Data Tables **75**, 1 (2000).

[76] R. H. Cyburt et al., The Astrophysical Journal Supplement Series **189**, 240 (2010).

[77] D. D. Clayton, *Principles of stellar evolution and nucleosynthesis: with a new preface*, University of Chicago Press ed (University of Chicago Press, Chicago, 1983).

[78] S. E. Woosley and R. D. Hoffman, The Astrophysical Journal **395**, 202 (1992).

[79] B. S. Meyer, T. D. Krishnan, and D. D. Clayton, The Astrophysical Journal **498**, 808 (1998).

[80] R. D. Hoffman, S. E. Woosley, and Y.-Z. Qian, The Astrophysical Journal **482**, 951 (1997).

[81] Y.-Z. Qian, Progress in Particle and Nuclear Physics **50**, 153 (2003).

[82] J. J. Cowan, F.-K. Thielemann, and J. W. Truran, Physics Reports **208**, 267 (1991).

[83] S. Wanajo, The Astrophysical Journal **666**, L77 (2007).

[84] A. Arcones and G. Martínez-Pinedo, Physical Review C **83**, 045809 (2011).

[85] C. Freiburghaus, S. Rosswog, and F.-K. Thielemann, The Astrophysical Journal **525**, L121 (1999).

[86] S. Rosswog et al., Monthly Notices of the Royal Astronomical Society **439**, 744 (2014).

[87] F. X. Timmes and D. Arnett, The Astrophysical Journal Supplement Series **125**, 277 (1999).

[88] F. X. Timmes and F. D. Swesty, The Astrophysical Journal Supplement Series **126**, 501 (2000).